TriAlign Technical Design

An Agent for Clinically Meaningful Comparison of Eligibility Criteria

Architecture, Dataset, Pair Semantics, Interface Concepts, and Research Notes

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Abstract

TriAlign compares clinical-trial eligibility rules using structured constraints, dense representations, and contextual reasoning. This document defines what constitutes a clinically meaningful criterion relationship, how source evidence and normalization are retained, how an agent selects and reconciles comparison tools, and how a clinician could inspect and correct a result. The frozen criterion release contains 3,537 records from 100 trials. The first development pair-review release preserves 75 decisions, including 67 clean comparisons and eight queued cases. A functional local comparison interface and bounded agent runtime are implemented. Fine-tuned models and independent clinical evaluation remain subsequent deliverables.

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1 Motivation

1.1 The Clinical Problem

An eligibility criterion determines what characteristics a patient must have to be included in a clinical trial. Two rules can reuse the same terminology while admitting different people, ie "age $\geq$ 18 years" vs "age $<$ 18 years" have similar vocabulary and disjoint allowed age ranges. Conversely, an abbreviation and an expanded term may express the same rule with limited word overlap. TriAlign is intended to pull seemingly similar criteria apart when clinical investigators, protocol designers, and research coordinators compare protocols.

The intended clinician-facing workflow compares the eligibility requirements of Trial A and Trial B. Criterion-to-criterion comparison is the computational unit underneath that view: retrieve rules about the same clinical concept, inspect their operational differences, and record an evidence-supported relationship. Individual-criterion search remains useful as a separate entry point. Potential applications include identifying precedent during protocol design, comparing amendments, documenting harmonization decisions, and inspecting whether superficially similar study populations are defined by materially different criteria. These applications motivate the system; their clinical benefit must be measured rather than assumed.

TriAlign does not determine an individual patient's eligibility, recommend a treatment, estimate recruitment yield without patient data, or establish causal transportability. A criterion-level comparison is not a claim that two complete protocols recruit the same population: their remaining rules, sites, ascertainment practices, and treatment settings may differ.

1.2 Trial comparison workflow

The clinician selects two trials, uses Modern AutoCrit to extract and structure their eligibility criteria, reviews the extracted records, and imports the resulting criteria files into TriAlign. TriAlign is intended to align comparable requirements and display their relationships, differences, source evidence, unresolved cases and requirements without a supported match. Figure 1 is the same diagram used in the progress presentation.

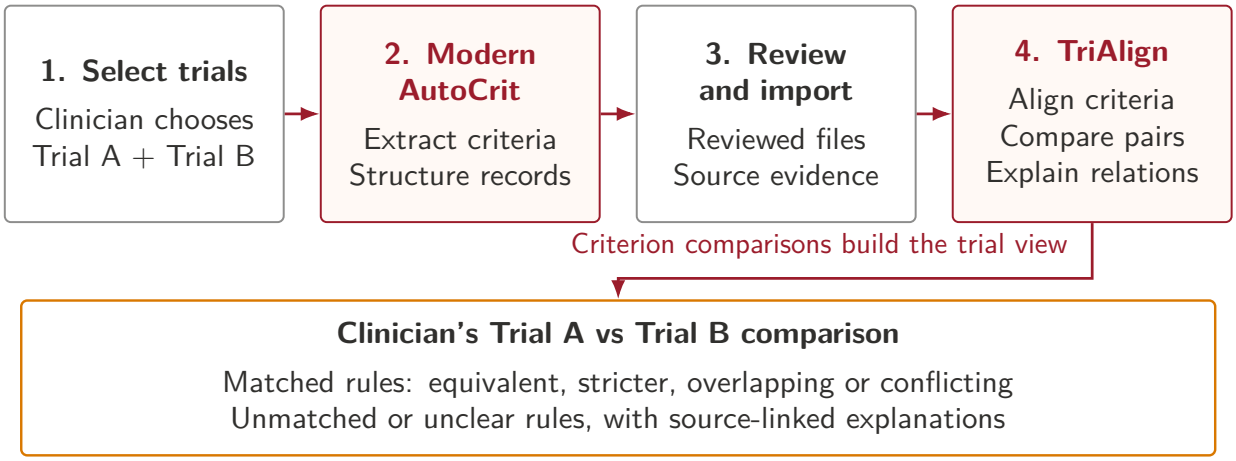


Figure 1: Intended workflow: a clinician compares two trials through source-grounded criterion comparisons. A relationship between two rules does not establish a relationship between the complete trial populations. Trial alignment and the comparison interface remain planned.

The encoder retrieves potentially corresponding criteria; the relationship component and contextual review determine how selected pairs compare. Criterion-pair labels train and evaluate those components. A trial view then assembles their results, with support for one-to-many or grouped matches where Boolean scope requires it. A retrieved neighbor is not automatically a semantic match, and no match found is not proof that a requirement is absent from the other protocol. This workflow does not yet define a validated single whole-trial similarity score. A future combined application could remove the manual file handoff while retaining the review checkpoint and separate component modes (Section A.7).

1.3 Why textual similarity is insufficient

The clinical meaning of a rule depends on more than its entity name.

Dimension	Illustrative difference	Comparison implication
Polarity	Require a condition versus exclude that condition.	Interpret the allowed population before using numerical or categorical agreement.
Threshold and boundary	Age ≥ 18 versus > 18.	A one-point boundary changes equivalence and containment.
Units and references	Absolute bilirubin concentration versus a multiple of ULN.	Equal numbers need not be comparable quantities.
Time and anchor	Event in the previous six months versus six months after treatment.	Duration, direction, and reference event must agree.
Repetition	A single observation versus two observations on separate occasions.	An identical threshold can encode a stronger evidentiary requirement.
Disease specificity	Active disease, previous disease, anatomical site, stage, or severity.	Ontological similarity alone does not establish equal eligibility.
Exceptions	Other malignancy excluded except specified cured cancers.	Removing the exception reverses the interpretation of an allowed subgroup.
Boolean scope	$A \wedge B$ versus $A \vee B$, possibly inside a larger clause.	Independent atoms must retain their branch conditions.

An embedding score can be useful for finding a likely comparison without being a valid clinical relationship score. In particular, cosine similarity is symmetric, whereas "A is more restrictive than B" is directional. A high cosine score therefore cannot, by itself, select a directional label or prove equivalence.

1.4 Research questions and hypotheses

Three questions we.re interested in are:

1. Which phenomena produce false-high similarity for clinically different rules, or false-low similarity for equivalent rules, in general and biomedical encoders?
2. Does supervision from reviewed positives, ordinary negatives, hard negatives, and ordered anchors improve held-out retrieval and relationship prediction?
3. Does selective tool routing improve evidence support, calibration, reviewer effort, latency, or cost relative to an LLM-only judge and an always-run fixed chain?

2 Eligibility Criteria Dataset

2.1 Modern AutoCrit and the upstream handoff

Modern AutoCrit produces the source records used by TriAlign. It accepts an NCT identifier, protocol text, or a protocol PDF; locates eligibility content; distinguishes inclusion from exclusion; extracts atomic constraints; and preserves the parent statement and local context. Extraction records include clinical fields, source identifiers, protocol versions, validation findings, and review status. TriAlign consumes reviewed outputs rather than replacing this extraction system. Criterion review corrects segmentation, polarity, measurement fields, normalization, and scope.

Release property	Verified value or source
Dataset version	`trial100_v1`
Source workbook	`data/trial100_v1.xlsx`, sheet `Human Review`
Frozen snapshot	`data/processed/trial100_v1.parquet`
Manifest	`data/processed/trial100_v1.parquet.manifest.json`
Retained criteria / trials	3,537 / 100
Review status	3,537 corrected; zero pending
Export readiness	All retained rows export-ready; zero recorded validation issues or readiness blockers
Schema version	`0.1.0`
Pair labels	Not included in the criterion release

Table 2: Current release facts are drawn from the frozen manifest. A review-status flag is distinct from a pair label or a calibrated confidence estimate.

2.2 Current frozen release and historical counts

The cohort design in the proposal is 50 oncology and 50 cardiovascular trials. Therapeutic-area evaluation will use an explicit trial-level mapping; that design statement is not a verified per-row therapeutic-area annotation in the manifest.

The initial extraction had 3,145 candidates. Splitting, correction, and removal changed the number of retained records. The October 20 proposal milestone reported approximately 2,900 reviewed cases and a remaining uncertainty queue. Those historical workload figures are not the denominators for v1 evaluation. Use the frozen manifest and saved split files for every reported result.

2.3 Record schema and evidence hierarchy

Field group	Representative fields	Technical use
Identity and lineage	`criterion_id`, `row_id`, `trial_id`, `parent_statement_id`, `origin_criterion_id`	Stable joins, split membership, split-parent lineage, and audit history.
Source evidence	`source`, `source_excerpt`, `parent_statement`, `surrounding_context`	Human inspection, evidence citations, and scope reconstruction.
Clinical predicate	`criterion_type`, `domain`, `entity`, `comparator`, `value`, `unit`, `value_kind`, bounds and inclusivity	Reconstruct the stated constraint before interpreting its eligibility effect.
Context and logic	`negated`, `temporal`, `repetition`, `qualifier`, `logical_operator`, `boolean_group_id`, atom indices/counts	Retain conditions and connections among atoms; flat fields alone do not certify a complete Boolean tree.
Normalization	Raw and normalized entity/value/unit fields; normalized bounds; `reference_type`, `reference_multiplier`; methods/status/confidence	Controlled matching while preserving source-grounded values and normalization provenance.
Protocol provenance	`source_uri`, `source_version`, `source_retrieved_at`, `protocol_hash`, `source_id`	Identify the exact source snapshot rather than silently following a changed registry page.

Field group	Representative fields	Technical use
Review and release	`review_status`, `export_ready`, issue/blocker lists, review-method metadata	Admission to the release and traceability of subsequent corrections.

The evidence hierarchy is the protocol wording and its governing context, followed by the reviewed structured interpretation and normalization. When an atom and its parent appear inconsistent, the system should expose the conflict and request adjudication. Opening a current registry link does not establish that its wording matches the version originally extracted.

2.4 Normalization and serialization

Terminology normalization supports retrieval across abbreviations, spelling variants, and aliases. It must preserve disease stage, severity, anatomical site, assay, and treatment context. A normalization dictionary should carry its own version and provenance, and proposed synonym merges should be evaluated for both false merges and missed aliases. A method/status field can describe normalization without establishing that its numeric confidence is calibrated.

Physical-unit normalization and reference normalization are different. "Bilirubin $\leq$ 1.5 $\times$ ULN" uses `reference_type=ULN` and multiplier 1.5. ULN is not a physical concentration unit. A local upper limit can vary by assay or institution, so reference-relative and absolute thresholds require explicit reference information before conversion. Likewise, systolic/diastolic notation does not authorize inferring an AND/OR relationship that the source did not specify.

The source's clinical and normalization fields are retained in the snapshot. Schema loading standardizes missing values, Boolean fields, issue lists, and the casing of selected categorical status fields. Because Parquet requires homogeneous column types, the mixed `value`, `value_raw`, and `value_normalized` columns use manifest-declared JSON scalar encoding. The project loader decodes these columns, preserving numeric values, categorical strings such as "001", and Booleans. Numerical bounds remain typed numeric columns. Direct consumers of Parquet must read the manifest; they must not interpret JSON scalar text as already-decoded clinical values.

2.5 Release integrity and changes after freezing

The release records the source and Parquet hashes, schema version, criterion-ID fingerprint, creation time, Git revision, and exporter/script hashes. The export was reopened through the project loader and reconciled record by record against source-derived records. These checks establish serialization integrity, not independent clinical certification of every interpretation.

A correction after freezing creates a new dataset version, an explicit change record, and a regenerated manifest. Existing pair records must continue to identify the corpus version against which they were labeled. A source edit must not silently change an already-published training or test example. Shared public-source subsets should retain this provenance and an intended-use statement. Sharing the criterion corpus with the group mentioned in the October presentation feedback must not expose protected TriAlign test labels.

2.6 Optional external data

Chia is an annotated eligibility-criteria corpus with entity, relation, and scope representations [15]. It may supply external terminology or logical-complexity examples. Its annotations do not directly provide TriAlign's eight pair relationships. Any use requires an explicit schema mapping, license/source review, overlap checks, and separate reporting; it must not be silently mixed into the v1 benchmark.

3 Clinical Pair Definitions

3.1 The ordered comparison unit

A pair contains two reviewed criteria, A and B, normally from different protocols, together with their corpus versions, trial/criterion identifiers, source excerpts, parent context, structured fields, and sampling provenance. The input order is part of the task. An unordered pair-group identifier can deduplicate cases and hold reversed views together, while an ordered pair identifier or orientation field preserves the direction of prediction. Mirrored cases must not become independent examples across train and test.

Criterion approval establishes that one record represents its source rule. It does not establish a relationship to any other criterion. Pair labeling is a second review with its own evidence, uncertainty, and adjudication record. The following formalization is an operational draft of the labeling contract; JM must approve the final guide before benchmark labels are frozen.

3.2 Allowed populations and comparability

Let U_γ be a common comparison universe under explicit context γ, including the measurement, population, temporal anchor, and branch scope. Let $P_c(x;\gamma)$ be the clinical predicate stated by criterion c, with its internal negation and exceptions already interpreted. The criterion-level allowed set is

$$S(c;\gamma) = \begin{cases} \{x \in U_\gamma : P_c(x;\gamma)\}, & c \text{ is an inclusion rule,} \\ U_\gamma \setminus \{x : P_c(x;\gamma)\}, & c \text{ is an exclusion rule.} \end{cases}$$

An internal negation is applied inside P_c; inclusion/exclusion is applied once outside it. A categorical value "absent" and a negation flag must not accidentally invert a predicate twice. Record their combined interpretation before comparing sets.

Set operations are meaningful only after establishing a common operational frame. Age and platelet count can both restrict a person, but this does not make their comparison a useful "partial overlap" label. First determine conceptual comparability. Related measurements such as serum creatinine and estimated glomerular filtration rate may require a clinical mapping that is not available; terminology similarity alone is insufficient.

3.3 Eight relationship labels

For comparable, interpretable criteria, write $S_A = S(A;\gamma)$ and $S_B = S(B;\gamma)$.

Label	Operational definition	Illustrative pair
Equivalent	$S_A = S_B$, including applicable scope and exceptions.	Inclusion age ≥ 18 versus age ≥ 216 months after a valid age-unit conversion.
A more restrictive	$S_A \subsetneq S_B$.	A: inclusion age ≥ 21; B: age ≥ 18.
B more restrictive	$S_B \subsetneq S_A$.	The preceding pair reversed.
Partial overlap	Nonempty intersection, with neither set containing the other.	Inclusion ages 18–65 versus 50–80, all endpoints inclusive.
Contradictory or disjoint	$S_A \cap S_B = \varnothing$ in the common operational frame.	Inclusion age ≥ 18 versus age < 18. This does not claim that the complete trials are mutually exclusive.
Related but not comparable	A meaningful clinical relationship exists, but a defensible shared constraint representation or mapping is unavailable.	Renal-function rules expressed through different measures without a supported conversion.

Label	Operational definition	Illustrative pair
Unrelated	No meaningful direct clinical comparison for this task.	An age threshold versus an unrelated laboratory requirement.
Ambiguous	Missing or conflicting evidence prevents a defensible choice among the relationships.	A truncated rule whose polarity or governing exception cannot be established.

The distinction between "related but not comparable" and "ambiguous" is important. In the first case, the concepts are understood but the relationship cannot be ordered in the available representation. In the second, the meaning or relevant context of the input itself is insufficient. Annotators should record the reason, not force uncertain cases into an ordinary negative class. Empty or internally inconsistent allowed sets also require an explicit inconsistency disposition rather than a misleading containment label based on the mathematical empty-set convention.

3.4 Labeling decision procedure and worked safeguards

1. Confirm the source version and that both records are eligible for pair review. Inspect parent statements and any linked atoms needed for scope.
2. Identify the clinical quantities, anatomical/clinical context, applicable population, measurement method, temporal window, and exceptions.
3. Classify clear unrelatedness, related non-comparability, or unresolved ambiguity. Otherwise establish the common comparison frame.
4. Translate each rule into its criterion-level allowed predicate. Validate units/references, boundary inclusivity, internal negation, and eligibility role.
5. Determine equality, proper containment, intersection, or separation. Record a concise evidence-supported explanation and any assumptions.
6. Reverse A and B as a consistency check. Only the two restrictive labels swap; symmetric labels remain unchanged.

For example, inclusion "age ≥ 18" and exclusion "age ≥ 18" are not equivalent: the latter allows ages below 18 under the simplified common age universe. Similarly, changing a negation flag does not prove disjointness unless the complete predicates are complements. The allowed set for $\neg(\text{age} \geq 65)$ overlaps the set for age ≥ 18. Different lookback windows can change the relationship even when entity, threshold, and units match. A single atom from $A \wedge B$ is not an independent exclusion unless its conjunction condition remains explicit.

3.5 Annotation record and review protocol

Annotation component	Required content
Identity	Ordered pair ID, unordered pair-group ID, corpus version, A/B criterion IDs and trial IDs.
Evidence	Source excerpts, parent/context references, source versions/links, and field-level differences used by the reviewer.
Sampling	Every candidate-generation method that selected the pair, retrieval model/version and rank/score, phenomenon tags, and natural/constructed flag.
Decision	Eight-class label, direction, rationale, assumptions, source-ambiguity reason, reviewer uncertainty, and review status.

Annotation component	Required content
Adjudication	Reviewer identity, initial independent decisions when available, final adjudicator, revision history, and unresolved disposition.
Evaluation control	Family/group IDs, split, annotation-guide version, duplicate/variant links, and whether the case was used for development or demonstration.

JM owns pair definitions, construction, labeling, adjudication, and evaluation changes. JG consumes the approved development records for software and model work and does not redefine gold labels to fit outputs. Independent second annotation of a documented subset is desirable; if a second reviewer is available, preserve the initial disagreement and JM's adjudication rather than reporting only the consensus. The guide should be piloted and revised before the held-out benchmark is locked. Model suggestions and similarity scores should be hidden during an independent gold-labeling round where feasible, reducing anchoring on the method being evaluated.

3.6 Family splits and pair sampling

Map trials to families before candidate generation. Related protocols, amendments, template-derived duplicates, and constructed variants must stay in one partition. A 70/15/15 family allocation is a starting configuration, not a guarantee of balanced criteria or relationship classes. Audit the actual family counts, oncology/cardiovascular composition, domains, and pair opportunities; record a fixed seed and the complete mapping. Missing split assignments must cause a pair-generation failure or explicit rejection, not silent inclusion in the known side's partition.

Candidate pools should combine canonical-entity matches, nearest neighbors from multiple encoders, lexical neighbors, random controls, and targeted clinical phenomena. Record overlapping sampling memberships instead of letting whichever generator ran first erase the others. Natural pairs measure behavior in the observed corpus; constructed paraphrases and controlled threshold/negation changes test diagnostic capabilities and are reported separately. Random controls mainly provide easy unrelated cases and must not dominate a benchmark intended to test subtle clinical differences.

3.7 Pair construction contract and development pilot v1

The operational annotation guide is contained in Appendix B, revised to `pair-guide-0.2` on November 7, 2026. The original pilot construction guide 0.1 is retained separately; decisions record the guide actually used rather than silently changing construction metadata. The guide applies the eight labels and comparison gates above to an ordered A/B example. Hash A then B for the ordered ID and sorted criterion IDs for the unordered group ID. Presentation direction is seeded independently of any presumed relationship. The pilot stores one direction per group; reverse-direction checks must exchange only the two restrictive labels. Predictions, random-control membership and structured differences never supply gold labels.

Development reservation before final splitting. Family mappings had not yet been audited when the pilot was constructed. The initial pilot therefore reserves twenty seed-selected trials for development rather than claiming a leakage-resistant train/test split. All reserved trials, and any related families discovered later, must be excluded from the final held-out test set. Merely inspecting a case while refining the guide counts as development exposure. The complete reservation list is saved in the pilot manifest. Audit family, template and duplicate links before freezing the remaining partitions and performing full-scale mining. The existing generic pair generator remains scaffolding; it is not the source of this pilot or an approved final split contract.

Sampling release. The original pilot used this generator command:

```
python scripts/build_pair_pilot.py
```

That builder is absent from the current checkout and must be restored before resampling. At construction it verified the frozen Parquet hash and checked unique IDs and reviewed statuses. Seed 20261104 selects the development pool and presentation order. The output is `data/processed/pair_pilot_v1/`; an existing directory is rejected instead of overwritten. Use `--output` to make a new version. The 75 natural cross-trial pairs comprise

25 same-entity pairs with structured differences, 25 further same-entity pairs, 15 lexical neighbors and 10 random controls. Literal entity matching normalizes case and whitespace only. Lexical sampling uses source-token Jaccard similarity of at least 0.25 within the same domain and ranks eligible candidates by that score. ASCII operators and numbers are retained as tokens; Unicode inequality-token support is a limitation of the initial heuristic. This is a sampling heuristic, not a clinical similarity score. Cap reuse at three pairs per criterion and four per unordered trial pair. Quota shortfalls cause failure rather than silent changes to the sampling design.

Store every qualifying method membership and the primary quota-filling method. Membership denotes eligibility under a rule, not independent selection or a known inclusion probability. The enriched convenience sample cannot estimate corpus-wide relationship prevalence. It contains no synthetic edits or embedding-mined candidates; add those as separately identified sampling conditions after the teammate's baseline integration is reviewed.

Review artifacts and acceptance. The JSONL packet stores full A/B records, source and parent evidence, available context and provenance, ordered/group IDs, corpus and guide versions, sampling information, and blank review fields. The labeling workbook separates review fields from source evidence. The retired Markdown packet is recoverable from the source archive in this document. The manifest records corpus, source-manifest and generator hashes, seed, development reservations, counts and output checksums. Verify 75 unique unordered pairs, cross-trial membership, quotas, reuse caps, empty labels and deterministic regeneration before review. These checks validate construction, not clinical correctness or label balance.

Keep the sampling release immutable. Save reviewer decisions in a separate versioned annotation file keyed by ordered ID, using the blank review object as its template. Require a supported common frame, cited evidence, rationale, assumptions, reviewer identity, low/medium/high reported uncertainty and status; an adjudicated record requires a final label. Preserve unresolved questions, independent initial decisions and adjudication history. Source corrections require a new corpus release and pair migration rather than edits to the frozen snapshot. Review the pilot before expanding the pool; revise the guide, audit family exclusions, and only then lock the held-out benchmark.

3.7.1 Guide revision and initial family audit

Guide 0.2 clarifies atom versus parent scope, inclusion/exclusion Boolean logic, conditional exceptions, measurement anchors, ULN conversions, consent capability versus completion, and the distinction between ambiguity and unavailable operational mappings. The original draft is preserved as a historical round. The current frozen development review is documented in Section C; reviewer identity and revision history remain explicit, and independent clinical validation is still a separate gate.

The November 7 family audit reconciles all 100 trials and screens all 4,950 trial pairs using archived eligibility and registry identity/lineage evidence. The 22 prioritized text-similarity pairs do not supply a supported in-cohort family merge. The map therefore has 100 provisional current families, not 100 proven independent studies. Three verified external links capture PREVAIL/CAP2, separately registered ZUMA-2 cohorts, and participant follow-up. Clinical-family membership and participant-overlap split groups are recorded separately. Full methods and source evidence are in Appendix D.

Unnamed linked studies for the master-screening trial NCT05419375 and rollover trial NCT07172919 remain unresolved. Retaining their conservative exclusions together with the pilot reservations excludes 21 current trials and 841 criteria from final held-out testing. No final partition assignments have been created. A scale-up split artifact must consume both `split_group_by_trial` and `holdout_excluded_trial_ids` from `data/processed/family_audit_v1/family_mapping.json`. The current generic split CLI does not enforce this map or reservations; integrate and validate that contract before mining a larger benchmark.

4 Agentic Architecture

4.1 Runtime workflow and tool contracts

The system has an offline data/model workflow and an online comparison workflow. Offline work freezes the corpus and splits, builds candidate pools, collects labels, trains models when justified, selects policies on development data, and prepares versioned indexes. Online inference uses frozen weights and versions; reviewer corrections enter an audit log and a later offline release rather than silently changing a running model.

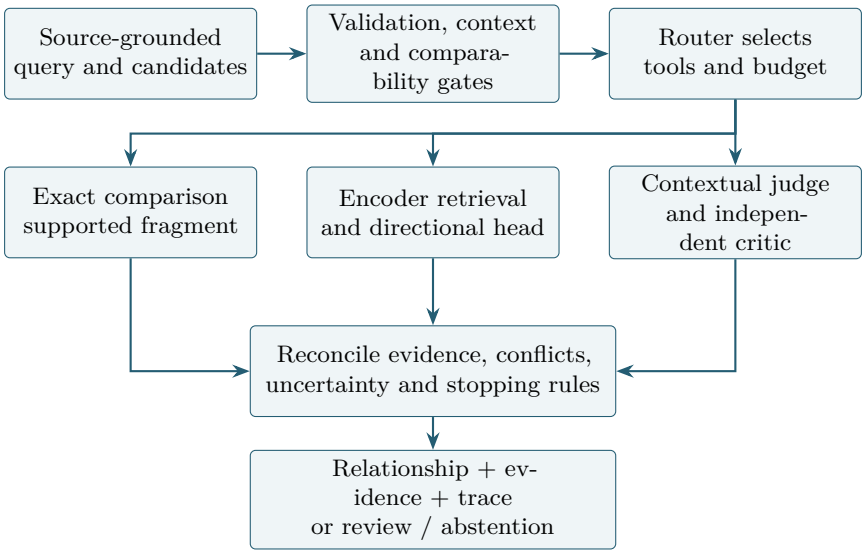


Figure 2: Proposed runtime. Retrieval proposes comparisons; routing decides which additional reasoning is required. A route can use several tools without making every request execute the same fixed chain.

The input contract contains the two records, necessary parent evidence, corpus/split identifiers, requested comparison mode, and a tool budget. A tool output should contain a supported label or unresolved result, explicit preconditions, field differences, cited evidence, assumptions, and a requires-context flag. Scores must identify their meaning: cosine similarity, uncalibrated classifier score, calibrated probability, or a deterministic logical result are different objects.

The initial router can be a transparent rule policy. Numerical records with matching supported quantity/context signatures may enter the exact route; large candidate searches use the encoder; exceptions, ambiguous history, measurement incompatibility, and unresolved scope invoke contextual review. A learned routing policy is a later experiment requiring its own labels and cost/quality evaluation, not a prerequisite for an auditable prototype.

4.2 Exact Comparisons

4.2.1 Supported fragment and admission gates

The exact route is a conservative comparator for an explicitly supported fragment: scalar thresholds, complete intervals, supported unit conversions, categorical predicates with controlled semantics, and simple temporal windows with shared anchors. More complex temporal expressions, clinical hierarchies, assay conversions, and nested Boolean conditions require additional representation or contextual reasoning.

Before emitting an exact relationship, check entity identity, measurement method, applicable population, units/reference basis, temporal anchor, repetition, exceptions, negation, and Boolean scope. Numerical validity alone does not satisfy these gates. Missing context is not a wildcard match. An unsupported case returns a reason for escalation rather than a fabricated exact answer.

4.2.2 Interval algebra and directional containment

Represent a numerical constraint by a lower endpoint, lower inclusivity, upper endpoint, and upper inclusivity. Missing endpoints denote unbounded directions only where the comparator actually establishes that meaning. For example, $x > t$ becomes $(t, +\infty)$ and $x \leq t$ becomes $(-\infty, t]$. Strict and inclusive boundaries remain distinct.

Compare the resulting allowed sets, including any complement introduced by an exclusion rule. First check equality, then proper containment, then disjointness, and finally genuine partial overlap. At a shared endpoint, intersection exists only if both intervals include that point. Excluding an interval can yield a union of two intervals; a comparator that supports only one interval must escalate that case or implement a verified interval-union representation. It must not drop one side of the complement.

An exact explanation should expose the transformation, for example: "A allows $[21, +\infty)$ and B allows $[18, +\infty)$; A is a proper subset of B." A witness such as age 19 can illustrate why B allows an additional case. Such a witness is a logical example, not an actual patient or an estimate of how many people a trial would enroll.

4.2.3 Units, reference ranges, and time

Use versioned, deterministic conversion rules only for commensurable quantities. Record both the source and converted value. An absolute concentration cannot be compared to a ULN multiple without a shared reference value and assay context. Drug washouts expressed in half-lives similarly require a drug-specific half-life and an approved interpretation before a fixed-day conversion is possible.

Temporal comparison needs the event, anchor, direction, duration, and inclusivity. "Within 30 days before first dose" differs from "30 days after the last dose." A rule using the lesser of 28 days or five half-lives is a conditional duration expression, not a single fixed bound. Endpoint tolerances and investigator exceptions should remain explicit. Repetition requirements need a compatible observation definition and spacing rule; an extra observation is not automatically a stricter criterion if its sampling or allowed timing differs.

4.2.4 Boolean scope, negation, and exceptions

Do not assume a flat list of atoms is a complete formula. In an exclusion $(A \wedge B) \vee C$, the record for A must remain conditional on B, and C remains an alternative branch. The v1 parent/context fields preserve evidence, but exact Boolean execution should require an explicit, reviewed expression tree or an equally precise scoped representation. If this is unavailable, use contextual review. An SMT solver is an optional future extension, not evidence that the current flat schema already supports arbitrary logic.

An exception is part of the governing rule. If a disqualifying malignancy predicate is M and an allowed cured-cancer exception is E, the exclusion predicate can be $M \wedge \neg E$ when that accurately reflects the source. Encoding E as an independent exclusion reverses its role. Internal negation, categorical absence, and exclusion type must be reconciled once, explicitly.

4.2.5 Current implementation boundary

The repository contains a starter interval comparator and deterministic-first orchestrator. They are scaffolding for this specification. The current comparator does not implement proper-subset direction, complete inclusion/exclusion set semantics, or all of the context/reference gates above. Its hard-coded confidence values are not calibrated probabilities. The exact-route acceptance suite should therefore include boundary, containment, complement, scope, unit, reference, and temporal cases before the route is described as clinically reliable.

4.3 The Encoder Model

4.3.1 Role, input representations, and inference

The encoder supports scalable candidate retrieval and a representation for supervised pair prediction. Given serialized criterion text $t(c)$, produce a unit vector

$$z_c = \frac{f_\theta(t(c))}{\|f_\theta(t(c))\|_2}, \qquad s(A, B) = z_A^\top z_B.$$

The text serializer is part of the experiment configuration. An initial fair comparison supplies identical raw criterion text to all baselines. Additional ablations can use a structured rendering or a source-plus-context rendering, but their results should be reported as different input conditions. Preserve comparators, units, negation, timing, reference basis, eligibility role, and necessary qualifiers in a structured rendering. Never include gold labels, reviewer rationales that reveal them, or held-out annotation metadata.

The planned general baseline is `sentence-transformers/all-mpnet-base-v2`; its model card describes a 768-dimensional sentence representation and default truncation beyond 384 word pieces [4]. The biomedical baseline is `FremyCompany/BioLORD-2023`, which builds on MPNet with definition/ontology-grounded biomedical training [5].

For each model, pin the revision and tokenizer, inspect truncation, use the specified pooling implementation, and save normalization and input settings. Do not silently omit a critical exception at the end of a long parent.

Retrieval excludes the query itself and, for the cross-trial task, its trial. Every experiment declares its permitted candidate partition. Cache keys include corpus hash, serializer version, model revision, and inference configuration. A cached embedding matrix is invalid when any of these changes.

4.3.2 Directional relationship head

A shared encoder with cosine similarity cannot distinguish A-more-restrictive from B-more-restrictive. A proposed ordered head uses

$$h_{AB} = [z_A; z_B; z_A - z_B; |z_A - z_B|; z_A \odot z_B; \phi(A, B)], \qquad p(y \mid A, B) = \mathrm{softmax}(g_\psi(h_{AB})),$$

where ϕ contains approved structured differences and missingness flags. This is a proposed feature design, not a selected final architecture. Compare a simple frozen-embedding classifier against encoder fine-tuning before adding complexity. Structural features must be derived from the input, not gold adjudications. A high classifier score does not authorize ignoring failed comparability gates.

4.3.3 Supervision and training objectives

Training data may contain equivalences, ordinary negatives, hard negatives, and ordered anchors. Hard negatives include lexical neighbors whose polarity, boundary, timing, or exception changes the relationship. They should not include unlabeled pairs arbitrarily declared negative. Restrictive and overlapping pairs remain conceptually related; treating every non-equivalent pair as a maximally distant negative would damage useful retrieval geometry.

One training option is a supervised contrastive loss on justified positives, an ordered classification loss, and a diagnostic ordering loss:

$$\mathcal{L} = \lambda_{\mathrm{con}}\mathcal{L}_{\mathrm{con}} + \lambda_{\mathrm{cls}}\mathcal{L}_{\mathrm{CE}} + \lambda_{\mathrm{ord}} \max\{0, m - s(a, p) + s(a, n)\} + \lambda_{\mathrm{swap}}\mathcal{L}_{\mathrm{swap}}.$$

Here (a, p, n) is an explicitly reviewed relative-similarity triple, not an ordering inferred from restrictiveness alone. The swap term penalizes a failure to reverse directional labels while preserving symmetric ones. Loss weights, margin, negative sampling, early stopping, and class weighting are selected on development data. Report learning curves and at least three training seeds for the final trained comparison. If supervision is inadequate or no improvement is observed, retain the strongest frozen model and report the result.

4.3.4 Retraining procedure and release checklist

Proposed design. Retraining is an offline maintenance operation, never part of a demonstration comparison. This checkout has encoder inference wrappers but no implemented training entry point. The following is the required runbook for that implementation; it does not describe an executable training command that already exists. The training deliverable must add exact preparation, training, evaluation, export and index-building commands to the README, together with a checked-in example configuration.

1. **Decide what needs updating.** Record the reason: new adjudicated pairs, a changed labeling contract, reproducible errors, or a new base model. Adding corpus criteria alone requires encoding/index refresh, not necessarily encoder retraining. Changing deterministic rules alone requires route validation. A label-contract change requires a versioned annotation migration and reevaluation of affected examples before training.

2. **Freeze the inputs.** Select an immutable corpus release and manifest, adjudicated ordered-pair labels, label mapping, serializer and family/group split manifest. Keep existing test families held out; group duplicates, templates and swapped pairs together. New examples enter the appropriate partition without moving old test cases into training. Save input hashes and validate IDs, label direction, missingness and leakage.

3. **Recreate the environment and configuration.** Save the code revision, dependency lock, hardware, base-model ID and full resolved commit. Specify frozen-head training versus encoder fine-tuning, feature schema, maximum sequence length, truncation audit, loss functions and weights, negative/triple sampling,

class weights, optimizer, learning rate, schedule, batch size, gradient accumulation, epochs or step limit, seeds, checkpoint frequency, stopping criterion and development selection metric. Record all effective values, including defaults. Resolve choices on development data.

4. **Estimate resources before the full run.** Run a short training smoke run and measure peak memory, step time and validation overhead on the chosen machine. A frozen-embedding head is the lower-cost first experiment; full encoder fine-tuning may benefit from a separate GPU machine. Reduce batch size or use validated gradient accumulation if memory is insufficient; record the effective batch and any change to negative sampling. Do not claim identical contrastive behavior merely because effective batch sizes match.

5. **Train and select without test feedback.** Generate only authorized training examples, fit learned preprocessing on training data, and save loss curves, development metrics and checkpoints. Select using the predeclared development policy. Save optimizer/scheduler state, random states and data-order configuration for resumable checkpoints; distinguish resuming an interrupted run from starting a new experiment. Repeat the final selected configuration with at least three seeds and retain failures as well as gains.

6. **Evaluate before promotion.** Compare the candidate with the current release and frozen baselines on identical partitions. Report retrieval and ordered-label metrics, false equivalence, swap consistency, clinical error slices, abstention/coverage, and local runtime and memory. Tune calibration or abstention only on development data. Run held-out test evaluation after configuration selection; do not repeatedly tune against it. If the candidate fails the declared quality or resource gates, retain the previous release and report the negative result.

7. **Export and rebuild compatible assets.** Export encoder, tokenizer, pooling/normalization configuration, relationship head, label map and serializer. Verify that reloaded outputs match the selected checkpoint within recorded tolerances. Rebuild corpus embeddings whenever encoder, serializer or corpus changes; a head-only change can reuse unchanged embeddings but still requires a new compatible application manifest.

8. **Release and rehearse.** Publish the approved artifacts and model card to the Hub, pin the resulting full commit, and retain the previous release for rollback. Package evaluation reports, configuration, seeds, hashes and index manifest. Download the release to the demonstration machine and verify offline startup, representative comparisons, latency and replay before selecting it as the default. Publication credentials stay outside the configuration and model package.

Training duration is not yet measured. Estimate it from the smoke run's steps per epoch, planned epochs and validation cost; report actual elapsed time and peak memory with the resulting release. Model training and demo inference may use different machines without requiring a hosted website.

4.3.5 Retrieval versus interpretation

A useful retrieval candidate can be an equivalent rule, a stricter rule, or a clinically important contradiction. Define the relevance policy for each retrieval experiment rather than assuming that only equivalent pairs matter. Training can pursue broad concept retrieval while the ordered head and exact tools distinguish clinical relationships. Evaluate both stages so a retrieval gain does not conceal a rise in false equivalence.

4.3.6 Hugging Face model release and local inference

Proposed design. Publish the trained TriAlign encoder as a versioned Hugging Face Model Hub repository. The Hub stores and distributes model checkpoints [2]; the demonstration downloads the selected release in advance and executes inference on the presenter's computer. The model repository identifier will be assigned when a trained release is ready. Publication is a release task, not a claim that a TriAlign checkpoint already exists.

The model release should contain encoder weights, tokenizer files, pooling and embedding-normalization settings, and the configuration needed to load the encoder. Package the ordered relationship head, label-to-index mapping, and approved structured-feature serializer alongside it when those components are trained. If the head has a separate artifact, pin both artifacts together in the application release manifest. The model card should identify the base checkpoint and its resolved revision, training-corpus and split versions, training configuration, evaluation

results and limitations, and applicable redistribution terms. Keep held-out labels and local API credentials out of the model package.

Pin the full resolved Hub commit for each encoder and relationship-head release. A moving branch name alone is insufficient to reproduce a run. Hugging Face supports revision-specific downloads and local caching [3]. During preparation, download all required model files, verify the selected revision and checksums, and run an inference warm-up. During the demonstration, load the prepared local model directory rather than resolving a newer remote revision. A missing or mismatched artifact should produce an actionable startup error instead of a silent download or model substitution.

Precompute corpus embeddings with the same model revision and serializer. At runtime, encode a new query locally, retrieve from the local index, and run the relationship head and exact tools on the same machine. CPU execution is the initial deployment target; locally available acceleration can be used after compatibility and latency are measured. Switching encoder releases requires regenerating the corpus index and validating its manifest.

This architecture supports model distribution through Hugging Face while keeping application execution local. A public website, a Hugging Face Space, and a dedicated inference endpoint are not required for the planned demonstrations. A hosted application can be evaluated as a later deployment option after the research workflow and operating requirements are established.

4.4 Fuzzy Clinical Reasoning: LLM Judge and Critic

4.4.1 Tasks and evidence-bounded prompts

The contextual route handles cases where an exact interpretation is not licensed by the supported fragment: implicit clinical language, exceptions, severity/history distinctions, incompatible measurements, treatment settings, or unresolved branch scope. The judge receives the ordered pair, source excerpts, necessary parent/context, reviewed structured fields, and any validated tool evidence. Retrieved neighboring criteria can provide precedent only when the experimental policy allows them; they cannot establish facts missing from the two source rules.

The prompt requests a structured answer: proposed label, direction, cited evidence spans, field/context differences, assumptions, uncertainty reason, and review/abstention disposition. Instructions distinguish absent evidence from evidence of absence. The model should name a missing temporal anchor or assay mapping rather than supply a clinically plausible one from memory. Source text is untrusted task data: embedded instructions in a protocol or retrieved document cannot authorize tool calls or override the task policy.

4.4.2 Independent criticism and bounded revision

The critic is a separate invocation that checks the proposed answer against the original evidence and any exact-tool certificate. Its checklist covers source support, inclusion/exclusion interpretation, negation, units and references, interval direction, timing, exceptions, branch scope, and unsupported assumptions. It also checks whether the label remains correct when the input order is reversed.

The critic may accept the supported result, identify specific repairable errors, or require abstention/reviewer escalation. One bounded revision followed by a final evidence check is an initial policy to evaluate; repeated self-discussion is not a substitute for missing evidence. A second call to the same model can have correlated errors, so critic agreement is not an independent human gold label or proof of correctness. Measure the critic's incremental benefit with a no-critic ablation.

4.4.3 Validation, reconciliation, and uncertainty

Validate schema, allowed label values, valid criterion IDs, required evidence, and quote grounding before accepting a model response. Invalid JSON, missing decisions, corrupted text, unsupported evidence, timeouts, and refusal/error responses are operational failures with explicit dispositions. They must not silently become unrelated or equivalent labels.

If the judge conflicts with the exact route, first recheck the exact route's preconditions. A mathematical result is conclusive only for its encoded predicate; the encoding itself may be wrong or incomplete. An unresolved conflict

triggers escalation. Do not reconcile incompatible labels by taking an unvalidated weighted average of cosine, a classifier score, and model self-reported confidence.

Separate *source ambiguity* from *model uncertainty*. The former concerns what can be established from the inputs; the latter concerns the system's reliability in choosing an answer. Confidence calibration requires held-out development labels. Until then, expose raw scores and categorical reasons for review. Select any automatic-accept threshold on development risk–coverage results, then freeze it for test evaluation.

4.5 Trace, stopping policy, and operational behavior

Each comparison receives a trace with ordered input IDs, corpus/source hashes, serializer and normalization versions, tool/model/prompt versions, route reasons, evidence, outputs, conflicts, final disposition, and human edits. Record wall-clock latency, token usage, configured pricing date when cost is reported, cache hits, retries, and tool errors. Store task evidence and auditable summaries, not a claim to expose private internal model reasoning.

The prototype should have explicit budgets for candidate count, tool calls, revision attempts, timeout, and cost. A valid exact result can stop without an LLM call; unresolved language can stop with abstention when the evidence is insufficient. Training and benchmarking must record these policies so reduced cost cannot be explained by silently dropping hard cases.

5 UI/UX Design

5.1 Clinical users and decisions to support

Design status: exploratory. The interface should be shaped by task observation and feedback from clinicians and trial staff. The following personas are design hypotheses rather than findings from a completed user study.

- A protocol investigator asks which historical rules operationalize the same concept and how a proposed change affects restrictiveness.
- A research coordinator needs a source-grounded explanation of a flagged difference and a way to send it for investigator review.
- A reviewer or methodologist needs to compare multiple protocols, record adjudications, and export an evidence trail for a design or harmonization meeting.

The primary interface question remains the proposal's practical task: given a current rule, which historical criteria are comparable, how do they differ, and what evidence supports the result? Avoid presenting the tool as a patient-screening interface when it has neither patient inputs nor validated screening logic.

5.2 Core interaction and information architecture

Proposed design. A three-region workspace can put a query and source context on the left, a ranked candidate list in the center, and a selected comparison on the right or in an expanded lower panel. A compact view should answer the clinical question before showing implementation traces.

1. Select a source trial and criterion, or enter a draft rule. If draft text is parsed, display the parsed interpretation for confirmation and mark it as an unreviewed draft rather than part of the frozen corpus.
2. Inspect the source wording, parent, role, population, time, and provenance.
3. Retrieve candidates with filters for therapeutic area, clinical concept, trial, source version, and comparison status. Clearly show which filters constrain the search; exclusion of a candidate is not a clinical decision.
4. Select a candidate and inspect its relationship, the decisive differences, supporting source spans, and any unresolved assumptions.
5. Accept, correct, or leave unresolved with a rationale. Distinguish a human review action from a system's automatic-accept recommendation.

6. Export the reviewed comparison with source and version identifiers.

A search result should show the candidate's source excerpt, trial, clinical concept, and relevance score type. A relationship label appears only when comparison has actually run. "No results" should distinguish restrictive filters, no candidate match, unavailable indexes, and tool failure. A blank or failed route must not look like a confident negative finding.

5.3 Visualization ideas retained from the proposal

These ideas remain candidates for the design, with implementation priority based on clinical usefulness and feasibility.

Concept	Useful presentation	Initial priority / limitation
Side-by-side constraint comparison	Align entity, role, comparator/value, units/reference, interval, timing, repetition, qualifiers, Boolean scope, and source; annotate the decisive difference.	Highest priority. Keep source language accessible beside normalized fields.
Numerical allowed-region plot	Draw the two admissible sets with open/closed endpoints, intersection, and containment; label the common quantity and units.	High for supported numerical cases. Draw allowed regions after polarity handling; hide the plot when context/conversion gates fail.
Agent decision trace	A compact route diagram expands to tool inputs, evidence, conflicts, critic disposition, and final action.	High. Show the real executed trace, not a decorative flowchart implying uncalled tools ran.
Method-agreement panel	Compare lexical/encoder rankings, exact result, judge, critic, and final decision without collapsing their distinct meanings.	High. Label cosine as similarity, not probability.
Uncertainty and review indicator	Show supported result, needs review, or unable to determine, with a specific reason and calibrated probability only when available.	Medium. Source ambiguity and model confidence are different dimensions.
Similarity versus clinical relationship plot	A cohort-level plot identifies false-high and false-low similarity by reviewed label and clinical phenomenon.	Medium, after labels exist. Appropriate for results/error analysis rather than the primary comparison screen.
Two-dimensional retrieval map	An optional projected neighborhood colored by reviewed relationship and marked by route.	Exploratory. Projection distances are not literal clinical distances and can distort the original geometry.

5.4 Additional clinician-oriented interface ideas

5.4.1 Protocol comparison matrix and review packet

A clinician may start with two protocols rather than one criterion. A comparison matrix could list clinical concepts as rows and protocols as columns, highlighting matched rules, missing matches, and unresolved scope. An unmatched rule means no supported match was found; it does not prove that the other full protocol lacks that requirement. A review packet could export selected comparisons, excerpts, source versions, reviewer rationale, and open questions for a protocol-design meeting.

5.4.2 Amendment difference view

For two versions of one protocol, separate wording-only changes from changed thresholds, windows, exceptions, or measurement methods. Show a proposed direction of change only for supported comparable rules. Preserve both versions and the amendment lineage. This is a useful future workflow, but amendments must remain in the same family for evaluation.

5.4.3 Source-first evidence drawer and context expansion

Every result can open an evidence drawer with highlighted source spans, parent statement, adjacent context, snapshot version, and registry link. Link a clinical summary such as "longer treatment washout" to the exact two windows that support it. Offer an explicit "context not sufficient" disposition instead of making the reviewer search for a hidden caveat.

5.4.4 What-if comparison sandbox

A draft workspace could let a user change one threshold or window and inspect the logical relationship to the original. Keep draft versions separate from the frozen corpus, and require confirmation of a parsed rule. A numerical witness can illustrate the set difference. Do not translate that difference into recruitment percentages or clinical safety claims without the additional data and validation those claims require.

5.4.5 Disagreement and adjudication queue

A review queue can prioritize exact/LLM conflicts, missing evidence, uncertain scope, or method disagreement. Show why each case was queued and support assigning a reviewer, adding an interpretation, and preserving earlier decisions. An override requires a rationale and evidence; a corrected result does not erase the model's original prediction from the evaluation record.

5.4.6 Saved cohorts of comparisons and precedents

Users may collect a set of source-backed precedents for a clinical concept, bookmark unresolved questions, and compare the same draft against several historical protocols. Group results by distinct trial/family so repeated template wording does not create a false impression of broad independent precedent. Saved sets should record filters and source versions for replay.

5.5 Accessibility, uncertainty, and usability validation

Use text and icons in addition to color; show full labels rather than only red/green agreement. Support readable typography, keyboard navigation, sortable lists, and expandable evidence without forcing a dense field table on every user. Prefer plain clinical explanations with technical details available on demand. Distinguish "evidence not available", "not comparable", "system error", and "review pending".

Evaluate prototype tasks with trial staff: find a supported precedent, identify a consequential threshold or scope difference, reject an unsupported equivalence, and complete an adjudication/export. Record task success, time, corrections, overlooked caveats, and users' interpretation of score labels. This is a proposed usability study; no clinician preference or efficiency gain is assumed. A within-person comparison with and without the tool should counterbalance task order if used to measure time savings.

5.6 Demonstration scope and resilience

For November 17, target the ranked list, aligned constraint view, supported interval plot, genuine route trace, and a method-agreement view. Add a confidence indicator only if its interpretation is defensible. By December 8, include the error-analysis plot and a documented uncertainty/review policy. The projected map and larger protocol matrix are optional extensions.

The presentation application starts on the presenter's computer and is displayed in a local browser. This is the planned demonstration deployment for both November 17 and December 8, avoiding recurring website and managed inference hosting costs. Model distribution follows Section 4.3.6; the runtime uses locally prepared weights, the frozen corpus, and a precomputed embedding index.

The live demonstration must include a success, an important failure, and a necessary abstention. Live local execution means the selected local tools actually run. Calling the external contextual LLM remains an optional online operation with its own API usage cost and network dependency. A local-only mode runs retrieval and supported exact/model comparisons, and explicitly escalates a case if it needs unavailable contextual reasoning. A cached mode replays saved versioned outputs, including saved judge/critic results where available, and is labeled as replay. A new input without a matching cached case cannot receive a fabricated replay result.

Before presentation, verify local startup, model/index revisions and checksums, inference warm-up, expected latency on the demonstration machine, source evidence, local review storage, and replay mode. Check credentials and connectivity only for the optional online route; stored source excerpts remain available locally even when registry links cannot be opened. The current local browser application implements the comparison workspace described below; larger visual analytics remain exploratory.

5.7 Implemented local agent runtime: November 7 update

Operational prototype. The application now compares two selected trials from the checksum-verified frozen corpus, or reviewed imported criterion files. It uses bidirectional TF–IDF neighbors with entity prioritization to construct candidate alignments. Rankings propose comparisons; they do not determine clinical relationships. The temporary encoder is explicitly identified as TF–IDF. A compatible encoder factory can replace it later; no trained relationship head is fabricated. MPNet/BioLORD adapters remain on the teammate branch pending her PR; the local runtime uses the original main-branch TF–IDF encoder.

1. The planner selects a versioned numeric, clinical-context or source-resolution skill from `src/trialign/agents/skills.yaml`. These are executable runtime instructions with tool permissions, not additional project Markdown.
2. A closed tool registry executes `inspect_frame`, `compare_constraints` and `cite_sources`. Each call records its arguments, version, outcome, timing and evidence hashes.
3. Input, before/after-tool, evidence and finalization hooks enforce the cross-trial boundary, source readiness, tool budget, cancellation and deadline. Literal source quotes must cover both sides. Known source defects route to source review rather than model repair of the frozen corpus.
4. Source-gated exact comparisons apply inclusion/exclusion role once, retain strict endpoints, support explicit unit conversions and discrete ECOG scores, and abstain on unsupported scope, timing, exceptions or source conflicts. The exact critic is a deterministic rule audit, clearly distinguished from an LLM.
5. Contextual cases invoke a structured-output judge followed by a critic. The orchestrator permits at most one revision, reserves budget for the critic, and publishes no clinical label after provider failure, budget exhaustion or critic veto. Operational failure is distinct from the clinical label ambiguous.

The provider uses the OpenAI Responses API with a strict JSON schema, `store=False`, bounded timeouts and no SDK retries. The adapter follows the official structured-output interface. The judge and critic are separate role invocations using the configured model; their agreement is not independent clinical validation. Instructions preserve the selected child as the annotation unit, exclude independent AND siblings, and retain governing OR groups. Quote validation establishes textual grounding, not semantic correctness. Context windows are bounded and may omit relevant remote text; the source view and human review remain necessary.

Default job limits are 16 candidate comparisons, eight model calls and 150 seconds. The interface permits bounded adjustments and cancellation; completed partial results survive interruption. A source snapshot, input hash, runtime source fingerprint, skill hash/version, requested and resolved model IDs, token usage, traces and append-only review history are saved atomically under `artifacts/demo/`. The directory is ignored by Git. Imported files are validated before persistence; the frozen corpus and pair release are never rewritten. Review saving requires a reviewer, rationale and current revision.

The interface implements trial selection, file import, live progress, candidate filtering, side-by-side evidence and parent context, rationale, critic feedback, executed traces, saved-run reload, review history and JSON export. Criterion coverage distinguishes a supported candidate, a case needing review, an unexamined case due to limits, and no retrieved candidate. None of these proves that a requirement is absent from the other trial. The browser is local; only the optional contextual route uses a remote service. Modern AutoCrit supplied implementation patterns for strict JSON boundaries, atomic storage and the local browser server. A combined extraction/comparison application remains future work rather than a feature of this prototype.

Verification and observed failure cases. The automated suite passes 18 runtime tests, covering polarity, endpoint/unit handling, source conflicts, critic veto, bounded revision, invented quotes, cancellation, atomic imports, and durable review history. A separate 12-case constructed offline harness passes via `pythonscripts/trialign.pyharness`. These are engineering diagnostics, not held-out clinical performance estimates. A headless browser check exercised import, comparison, rule-critic trace, review save/reload and JSON export, with no JavaScript errors; desktop and mobile layouts were inspected. A bounded live contextual run used four calls to `gpt-4o-mini-2024-07-18`, including a critic-requested revision. An earlier live run exhausted its revision allowance and withheld its label. The revised run obtained critic acceptance, but its explanation still illustrates possible contamination by neighboring consent/compliance clauses. Treat both as workflow checks requiring human interpretation, not validated predictions. No latency distribution, clinical accuracy, independent reviewer agreement or critic benefit has yet been measured.

5.8 Local startup and demonstration assets

Install the application dependencies and launch from the repository root:

```
python -m pip install -e ".[app,dev]"
python scripts/trialign.py serve --open-browser
```

Open http://127.0.0.1:8766. The loopback-only HTTP server serves the local browser interface and JSON endpoints; no public website is required. `app/review_app.py` is a compatibility launcher for this application. Set `OPENAI_API_KEY` and optionally `TRIALIGN_JUDGE_MODEL` for contextual reasoning. Environment variables override an explicit `TRIALIGN_ENV_FILE`, otherwise the repository's `.env`. TriAlign does not read configuration from Modern AutoCrit and requires no Modern AutoCrit checkout, server or installed package. The frozen corpus and source inventory are bundled within this repository. Compatible imported criteria may come from another extractor. A relocated-checkout regression loads all 100 trials and runs an offline comparison without an upstream directory or credentials. On a new checkout, copy `.env.example` to `.env` and provide a key. Credentials are read locally, never copied into tracked files or run exports. Turning off contextual reasoning leaves source-dependent comparisons unresolved.

Local asset	Preparation and runtime requirement
Python environment	Install and pin dependencies before the presentation; record the application Git revision and machine configuration.
Model files	Download the pinned Hub release in advance; retain weights, tokenizer, configuration, relationship head and checksums on disk.
Corpus and search index	Load the frozen Parquet/manifest and matching precomputed embeddings/index. Check corpus hash, model revision and serializer version together.
Demonstration cases	Save selected inputs, source excerpts, expected routes, output versions and representative success/failure/abstention cases.
Replay bundle	Keep complete saved outputs and traces locally; require matching input and configuration identifiers before replay.
Review records	Persist corrections, rationale and traces to a local SQLite database or versioned local files; preserve them across application restarts.

Local asset	Preparation and runtime requirement
Optional LLM route	Read credentials from the local environment, enable online calls explicitly, and set call/time/cost limits. Without that route, use local-only execution or clearly labeled replay.

The startup workflow should distinguish a valid warm cache from missing or incompatible assets. Rehearse from a fresh application process after all downloads are complete, including an offline run. The offline acceptance criterion is that the UI starts, local retrieval and supported comparisons work, local evidence is readable, and prepared cases can be replayed without network access. Arbitrary new fuzzy comparisons still require a running contextual model or escalation. Local startup removes the need for website hosting; it does not imply that training, initial model downloads, or optional external LLM inference have no resource or usage costs.

5.9 Compute requirements and demonstration latency

Proposed design. The initial goal is an interactive single-user comparison on the presenter's machine. No end-to-end latency measurements exist yet. A trained encoder performs inference rather than retraining for each input. With a loaded model and precomputed corpus embeddings, a new query needs one encoding, local retrieval, and comparisons for a bounded candidate set. Comparing two new criteria needs their encodings; comparing stored criteria can reuse compatible cached vectors. Deterministic checks do not generate text and are expected to contribute less delay than contextual LLM reasoning. These are architectural expectations, not measured performance claims.

For the current 3,537-criterion corpus, a 768-dimensional float32 embedding matrix occupies approximately 10.4 MiB before metadata/index overhead. An exhaustive query scan evaluates about 2.7 million vector components, which makes simple local vector search a reasonable initial design. Model weights, activations, tokenization and application overhead consume additional memory. Do not recompute the whole corpus or enumerate all criterion pairs in response to a single UI action. A full protocol comparison is a different workload: batch its query encodings and bound the candidates and contextual calls per criterion.

The main latency risks are cold model loading, long inputs, repeated encodings, and the contextual judge/critic route. An external LLM introduces network and service latency; sequential judge and critic calls add to the critical path. A local generative LLM may instead be limited by RAM, available accelerator memory and token generation speed, and can compete with the encoder for resources. Running the encoder locally does not establish that the complete generative route will be fast or fit on the same laptop. Backend, device and batch-size optimizations must be benchmarked on the actual machine rather than assumed to improve performance [1].

Keep the encoder and index loaded across UI interactions, warm them before presenting, reuse compatible embeddings and bound the retrieved top-k. Invoke contextual reasoning only when the route requires it, with explicit call, output-token, timeout and revision limits. Show retrieved evidence and completed exact results while contextual work is pending. On timeout, retain those results and display an unresolved state; offer a matching, explicitly labeled replay when available. Do not silently skip a required critic or present an incomplete relationship as final to meet a speed target.

Benchmark and acceptance procedure. Record machine model, CPU, RAM, accelerator, software/model/index revisions, input lengths and top-k. Measure cold startup separately from warm comparisons. Use representative exact, encoder-only and judge-plus-critic cases, including long inputs and failures; report sample counts, median and 95th-percentile latency, peak memory, cache status, route/call counts and external API time. Measure stage times for encoding, search, exact/head inference, judge, critic and rendering as well as the user-visible total. A proposed usability target is a warm local retrieval/exact result within 2 seconds and a completed contextual comparison within 30 seconds. These are targets to validate and revise, not promised runtimes. If the contextual route misses the budget, preserve the fast local workflow and use a budgeted external model, suitable local acceleration, or clearly labeled prepared replay for the demonstration.

6 Evaluation and Comparisons

6.1 Benchmark construction and evaluation units

Maintain separate pair-classification and retrieval evaluations. Pair classification tests a fixed set of ordered comparisons with reviewed labels. Retrieval tests a query and a declared pool of candidate criteria. A model can classify supplied pairs correctly while failing to retrieve them, or retrieve relevant concepts while confusing their relationship. Report both stages.

Use family splits before mining, preserve reversed/constructed variants in their source partition, and keep test labels unavailable to training, prompt design, model selection, agent memory, or retrieval of labeled precedents. Encoding held-out candidate text for a declared test search index is different from training on it; document the intended inductive evaluation. Fit TF–IDF vocabulary/IDF and all learned preprocessing on training text only. Select hyperparameters, calibration, and acceptance thresholds on development data. Run the untouched test evaluation after those choices are frozen.

Pool candidate judgments across the proposed baselines so one encoder does not determine the entire benchmark. Report judgment coverage, label counts, sampling strata, and the number of queries with known relevant candidates. Unjudged items are not automatically negatives. If the candidate pool is incompletely judged, call recall relative to that judged pool and disclose that it is not exhaustive corpus-wide recall.

6.2 Baseline Embedding Comparisons

Baseline	Fixed implementation / role	Key controls
Lexical	TF–IDF with lowercase word unigrams/bigrams and cosine ranking.	Fit only on training data; retain operators/negation in an audited serializer and record tokenization limitations.
General encoder	Frozen `sentence-transformers/all-mpnet-base-v2`.	Pin revision, pooling, truncation and input rendering.
Biomedical encoder	Frozen `FremyCompany/BioLORD-2023`.	Same candidate pool and input condition; document model-card license requirements before redistribution.
CReSE-informed representation	Eligibility-oriented representation/rephrasing approach, conditional on reproducible artifacts.	Distinguish an actual reproduced checkpoint from a newly implemented inspired baseline.
TriAlign encoder	Fine-tuned representation plus an ordered prediction head, when sufficient labels exist.	Same held-out evaluation; compare with frozen features and report training seeds.

The raw-text baseline is the simplest matched input condition. A second condition may provide the same structured/context rendering to every method. Do not compare a context-rich TriAlign input against context-deprived baselines without separately identifying that input advantage. A separate relationship classifier can be trained on each frozen embedding under the same protocol to isolate representation quality from the value of fine-tuning.

The model inventory in configuration also lists MiniLM, BGE, and SapBERT. They are optional candidates, not mandatory experiments or established improvements. Add one only with a documented question, pinned revision, supported input policy, and sufficient time to run the same evaluation. The required initial comparison remains TF–IDF, MPNet, and BioLORD.

6.3 Existing Eligibility Systems and Products: Research Notes

The following entries are research notes for the systems named in the proposal. Most are research methods or software artifacts, not interchangeable commercial clinical products. No shared eight-class benchmark results or TriAlign adapters are claimed here. Availability of a paper or repository does not establish installation success, compatible licenses, usable model weights, or reproduction of the published results.

6.3.1 CReSE: eligibility recommendation and representation

Research note; adaptation not yet implemented or evaluated. The published work introduces eligibility-criterion recommendation from trial information and an evaluation framework using contrastive learning and LLM rephrasing [7]. Its representation objective makes it a relevant comparator for eligibility-specific embeddings; its native task is not TriAlign's directional eight-label comparison.

Proposed comparison. If a suitable checkpoint or reproducible training path is available, evaluate criterion embeddings on the same retrieval pools, with any relationship head trained only on TriAlign training pairs. Otherwise report a clearly named CReSE-inspired experiment rather than claiming to reproduce CReSE.

Open research items. Identify code/checkpoints, exact preprocessing, data access and license, and possible source-trial overlap. Determine whether rephrasing preserves strict comparators, exclusion roles, and exceptions. Exit artifact: an availability/reproduction note and, only if feasible, a matched retrieval result.

6.3.2 TrialGPT: patient-to-trial matching

Research note; adaptation not yet implemented or evaluated. The official repository describes retrieval, criterion-level patient eligibility matching, and trial ranking, and provides code and data [9]. These inputs and outputs differ from comparing two protocol criteria. Its original patient-matching labels cannot be relabeled as TriAlign relationships without a justified task transformation.

Proposed comparison. Inspect its evidence-and-judgment prompt design and implement a separately named criterion-pair adaptation if defensible. Hold the supplied source evidence, base LLM, and allowed call budget constant when isolating prompt/workflow differences. Compare evidence support, abstention, latency, and cost on shared cases; preserve native results as background rather than importing their accuracy into a TriAlign leaderboard.

Open research items. Pin a repository revision, identify reusable prompts and output schemas, document all changes, and determine whether a pair adaptation is substantive enough to warrant inclusion. The cited clinical study remains background for the original task [8].

6.3.3 EligMeta: eligibility-aware evidence synthesis

Research note; adaptation not yet implemented or evaluated. EligMeta connects trial discovery and eligibility-aware meta-analysis. The paper separates LLM-generated rules/parsing from deterministic logic, weighting, and statistical pooling [10]. Its relevant overlap is eligibility alignment and reproducible execution; TriAlign does not implement its pooled-outcome estimand.

Proposed comparison. Investigate whether an exposed alignment component can operate on the common source-grounded cases. If not, compare architectural choices and a documented adaptation of its rule/parse/execute pattern. Never compare TriAlign pair macro-F1 directly to meta-analysis effect estimates or literature-selection recovery.

Open research items. Verify code and artifact access, alignment granularity, treatment of exceptions and absent fields, and compatibility of its similarity output with categorical relationships. Determine a fair matched-case input/output contract before implementing an adapter.

6.3.4 SatIR: constraint-satisfaction retrieval

Research note; adaptation not yet implemented or evaluated. SatIR formulates trial matching using explicit SMT constraints and scalable relational-algebra retrieval. The inspected paper is arXiv version 4 and links an

implementation [11, 13]. Its native problem is profile-to-trial retrieval rather than ordered criterion-pair labeling.

Proposed comparison. Study whether its constraint representation can support a restricted, reviewed pair-comparison fragment. Equality, containment, and separation can be investigated through satisfiability of set-difference/intersection formulas, provided both predicates and a common universe are faithfully encoded. A solver's answer does not validate the LLM translation that produced the formula.

Open research items. Record ontology assumptions, missing-data policies, supported temporal/Boolean expressions, source translation errors, and installation dependencies. Report unsupported coverage together with correctness on supported cases. Keep retrieval-system reproduction separate from a newly written logical pair adapter.

6.3.5 VERDICT: accountable decisions and rationales

Research note; adaptation not yet implemented or evaluated. VERDICT translates constraints and decision policies into SMT and uses SMT/MaxSMT reasoning to support explicit assumptions and pivotal conditions. It evaluates groundedness, consistency, and counterfactual self-faithfulness in clinical-trial matching [12]. The inspected paper links code/prompts in the Stanford OVAL repository [13].

Proposed comparison. Adapt its rationale-oriented questions: which source facts support the relationship, which assumptions are required, and what supported change would alter the decision? Test whether a stated pivotal condition actually changes a recomputed result. Treat these as additional audits of TriAlign explanations, not automatic proof that an explanation is clinically correct.

Open research items. Identify reusable policy/solver interfaces, the boundary between facts and assumptions, coverage of conditional exceptions, and the cost of faithful translation. A common-case adapter must declare every modification to its native decision task.

6.3.6 EC-RAFT: retrieval-augmented eligibility generation

Research note; adaptation not yet implemented or evaluated. EC-RAFT generates eligibility criteria from trial titles and descriptions using retrieval-augmented fine-tuning. Its paper identifies Llama-3.1-8B-Instruct as the base model and links released code [14]. Generation quality is a different task from pair relationship prediction.

Proposed use. Treat EC-RAFT as adjacent authoring research. It may inform a future draft-criterion workflow or supply separately marked constructed examples after source/clinical review. Generated criteria are not automatically valid training labels or clinical precedents.

Open research items. Check the generation/evaluation interface, retrieval-corpus overlap, and preservation of numerical and exception scope. Report any generation experiment separately from the main retrieval and relationship benchmark. It should not displace the mandatory matched-task baselines.

6.3.7 Common research-note template and comparison decision

For each system, maintain primary citation, inspected version/date, native task, input/output schema, artifact/license status, reproducible execution status, proposed adapter, task mismatch, expected cost, and comparison decision. Classify the outcome as: direct matched-task baseline; explicitly adapted common-case experiment; architectural/failure-mode comparison; or out of scope. Commercial products can be added later with the same template if the team identifies a specific accessible product and evaluation pathway. Do not invent a capability matrix from marketing claims or compare unavailable systems using fabricated results.

6.4 Agent comparisons and ablations

The main agent experiment compares an LLM-only judge, an always-run fixed chain, and routed TriAlign on the same supplied pairs, evidence, model versions, and output contract. Include an exact-only route with supported-case coverage where useful. Report both accuracy and expenditure; an expensive always-run system can improve accuracy while being less efficient.

Ablations isolate hard negatives, human corrections, normalization/context input, structured features/checks, critic calls, and targeted sampling. Change one component at a time where possible and state interactions when

several components change. Routing evaluation should include selection errors, missed opportunities to stop exactly, unnecessary LLM calls, and unresolved conflicts. Do not score an abstaining exact-only system solely on its easy resolved subset and compare that number to full-coverage models.

6.5 Metrics, uncertainty, and statistical reporting

Evaluation target	Measures	Interpretation controls
Retrieval	Precision/recall@k, MRR, equivalence-specific retrieval, reviewed anchor ordering.	State relevance definition, candidate/judgment pool, query denominator and judgment coverage.
Pair relationships	Eight-class macro-F1, per-class precision/recall, confusion matrix, directional F1 and swap consistency.	Keep class imbalance and asymmetric labels visible; report abstentions separately.
Clinical phenomena	Error counts/rates for thresholds, negation, units/references, timing, repetition, history, exceptions and compounds.	Give numerator/denominator and account for multi-tag cases.
Uncertainty	Calibration diagrams, Brier score where probabilities are defined, risk–coverage, inappropriate automatic acceptance, escalation/abstention errors.	Calibrate on development data; model self-report is not validated probability.
Evidence and explanation	Supported-claim rate, missing-condition errors, source-span support, contradiction and pivotal-condition checks.	Correct label and grounded rationale are distinct outcomes.
Workflow	Reviewer time and corrections, end-to-end latency, tool/LLM calls, tokens, cost, cache hits, failures.	Use matched tasks and comparable budgets; show live and cached execution separately.

At automatic-accept threshold t, define coverage as the fraction of all evaluated cases accepted automatically and selective risk as the error rate among those accepted. Report how many ambiguous cases are appropriately escalated and how many answerable cases are unnecessarily deferred. Uncertainty estimates should preserve trial-family dependence. Use paired method comparisons and family-aware resampling; a cross-trial pair touches two families, so an ordinary independent-pair bootstrap is not sufficient. One option is to resample held-out families and retain/weight pairs through their endpoint-family memberships, applying the same resample to all methods. Predeclare the chosen scheme and summarize sensitivity rather than implying that one simple bootstrap removes every dependency.

Report at least three seeds for trained models, effect sizes and intervals, sample sizes, family counts, and the limits of a 100-trial convenience cohort. Separate natural from constructed challenges. The final report must show representative false-high/false-low similarity, a routing or interpretation failure, and a necessary abstention alongside successes.

A Implementation, Delivery Milestones, and Reproducibility

A.1 Repository boundary and work ownership

The repository currently provides criterion/pair schemas, workbook and Parquet adapters, candidate generation, family-splitting utilities, a TF–IDF runner, a lazy Sentence Transformers wrapper, a starter exact comparator, a deterministic-first orchestrator, and a functional local browser review application (runtime update below). These interfaces do not establish that fine-tuning, contextual judging, calibration, full family-aware command-line splitting, or the proposed UI have been implemented. The pair generator's unordered ID and the split/comparator integration must be reconciled with this specification before benchmark production.

Workstream	Jonathan Ma (JM)	Jiatong Gao (JG)
Data and benchmark	Source release, normalization, family mapping, pair guide, construction, labels, adjudication, source validation and manifests.	Consume frozen development inputs and report interface/data issues without rewriting gold labels.
Models and agent	Define comparison semantics, metrics, and case-level evaluation requirements.	Implement and run TF–IDF/MPNet/BioLORD, routing/tools/critic, traces, comparison adapters where feasible, and UI-facing outputs.
Evaluation	Freeze splits/labels, audit directionality and reported case evidence.	Save predictions, rankings, runtime/cost logs, ablations and model/configuration artifacts.
Submission	Data/evaluation documentation, test-case index, provenance and contribution record.	Baseline/agent implementation documentation, reproducible execution, prompts, traces and UI assets.

Shared work includes reviewing changes, agreeing on contracts, inspecting successes and failures, rehearsing demonstrations, integrating the report, and recording contributions by artifacts and commits. JM retains authority over gold labels and evaluation-set changes; JG owns baseline and orchestrator implementation. Proposed independent second annotation should be arranged without confusing an annotator's input with final adjudication authority.

A.2 Milestone 1: October 20 presentation, completed

The first presentation introduced the clinical problem, source-data pipeline, 100-trial cohort, structured comparison, embeddings, agent design, and initial review work. Recorded feedback requested eligibility-focused system comparisons and consideration of sharing a documented criterion subset with another clinical-trials group working with JEV. Any such sharing preserves public source provenance, intended use, and evaluation-label separation.

A.3 Milestone 2: November 17 presentation

The format is 15 minutes plus 5 minutes of Q&A. The presentation must cover objective, dataset, workflow, preliminary positive and negative results, a live UI, limitations, and next steps.

1. JM delivers the frozen criterion release and data dictionary, approves the pair-labeling contract, establishes family splits before pairing, and provides a versioned pilot with evidence and directional consistency checks.
2. JG runs the three required frozen baselines on common inputs, prepares TrialGPT/EligMeta capability and adaptation notes, implements an auditable prototype router, and compares it with LLM-only/fixed-chain execution on the supplied development pairs.
3. Together, integrate ranked candidates, source/context, aligned constraints, supported numerical plots, real route traces, and cached replay. Save at least one success and one meaningful failure; demonstrate abstention when needed. Launch the application locally using Section 5.8, with all required model and index assets prepared in advance. Verify startup and replay without an internet connection.

The v1 criterion freeze is complete. Pair labels and implementation/evaluation artifacts remain subsequent deliverables. An approximately 200–300-pair pilot is an optional planning target, not an existing benchmark or a guaranteed class-balanced sample. If fine-tuning is not ready, report frozen baselines rather than an unvalidated training run.

A.4 Milestone 3: December 8 final presentation

The format is 20 minutes total for presentation, demonstration, and Q&A. The final package for this milestone includes the frozen corpus/labels and family splits; reproducible baseline and agent configurations; trained-model results if justified; comparative evaluation, ablations and uncertainty analysis; documented positive/negative cases; and stable local live/replay demos.

JM finalizes and validates the benchmark and all denominators. JG completes baseline runs, agent routing/reconciliation/critic/escalation, feasible external-system adapters, and UI integration. Freeze dataset, model, prompt, normalization, routing and calibration versions before the untouched test run. Do not change gold labels to accommodate method errors; genuine source corrections require a transparent versioned amendment.

A.5 Milestone 4: December 22 report and submission

The report is 5–8 pages excluding references and permitted appendices. This technical document is supporting material and is not constrained to that report length. The report should cover the clinical objective; source/final data and normalization; AI/agent workflow; benchmark and statistical methods; results and important failures; limitations and future improvements; and individual contributions.

The accompanying release should contain source data or a reconstruction manifest; criterion/pair manifests and family/split files; labeling guide and adjudication provenance; code and prompts; model/index/configuration versions; per-case inputs, predictions, evidence, traces and review outcomes; summary tables generated from saved outputs; and reproducible training instructions or model artifacts. Include the Hub repository identifiers and full resolved release revisions, a local model/index manifest, hardware and latency notes, and instructions for preparing an offline demonstration bundle. The README must cover installation, dependencies, API access for the optional online route, local startup, expected outputs, replay, reproduction and troubleshooting.

Verify a clean-environment execution of data loading, baseline inference, agent comparison, evaluation and local demonstration. Rehearse local-only, optional online LLM, and cached replay modes separately, including restart persistence of reviewer decisions. Link every report example to its frozen case and source evidence. Record cache behavior and the limits of reproducing stochastic external-model outputs. Freeze a tagged submission archive after the contribution statements and numerical claims are audited.

A.6 Deferred research direction: comparison across extraction granularities

Status: proposed research to revisit after progress-presentation feedback. This is not implemented in the current prototype and is not an established novelty claim. Retain the current criterion-pair demonstration and frozen release. Do not expand scope, regenerate labels or begin training for this extension before reviewing feedback and its feasibility.

Candidate contribution. Investigate group-level clinical requirement alignment whose conclusions remain stable when an extractor splits or merges a requirement without changing its meaning. Two protocols can express the same eligibility rule using different numbers of rows, Boolean groups or inclusion/exclusion forms. For defined age and ECOG measurements in a shared frame, exclusion of "age below 18 OR ECOG above 2" permits the same states as inclusion of "age at least 18 AND ECOG at most 2." Comparing isolated rows can miss this correspondence. Merely adding parent text to an LLM prompt is not the proposed methodological contribution.

1. **Group alignment method.** Represent criteria with explicit Boolean, exception and governing-context relationships. Identify supported one-to-many and many-to-many correspondences, then compare the smallest semantically complete groups. Define the supported logic fragment and common measurement frame. A meaning-preserving split, merge or logical rewrite should leave the group relationship unchanged; an eligibility-changing edit should change it when relevant.
2. **Interpretation-dependent results.** Retain multiple source-supported interpretations when attachment or scope is unresolved. Report a stable relationship only when it holds across the retained interpretations. Otherwise identify the alternative relationships and the specific source clarification that distinguishes them. This is not a guarantee covering every possible clinical reading. Solver correctness does not establish that the source was translated faithfully.

3. **Learning objective.** Explore joint group alignment and ordered relationship prediction, with consistency across reviewed, meaning-preserving decompositions. Keep counterpart relevance distinct from population equivalence: contradictory age rules may be important counterparts to retrieve. Do not push all contradictory pairs apart indiscriminately. Neither a fine-tuned encoder nor an added critic alone establishes novelty.

Evidence needed. Extend annotation with separately versioned group identities, governing scope, source evidence and any admissible interpretation alternatives. Do not silently change the annotation unit or overwrite the frozen pair contract. Construct a reviewed evaluation set containing natural group alignments, controlled meaning-preserving decompositions and genuinely eligibility-changing edits. Keep constructed examples distinct from natural cases and all transformations of a source family within one partition. Measure group alignment, directional relationships, invariance to valid restructuring, sensitivity to semantic changes, coverage/abstention and cost. An always-constant predictor must fail the sensitivity component. Compare structured matching, lexical/dense retrieval, LLM-only judgments and the current routed system; ablate group structure and interpretation handling. The existing 75-case development release is not, by itself, evidence supporting the proposed group-level method.

Novelty and decision gate. The hypothesis is that explicitly handling different extraction granularities and interpretation uncertainty improves cross-trial comparison over isolated criterion matching. Validate the distinction against eligibility representation learning, formal constraint reasoning and amendment analysis, including CReSE, EligMeta, SatIR, VERDICT and AMEND++ [7, 10, 11, 12, 6]. A targeted search that finds no identical system does not prove priority. After presentation feedback, decide whether to pursue a bounded numerical/Boolean prototype, seek additional clinical annotation, or retain the narrower current project. The current UI, agent runtime and source-preserving imports provide infrastructure for this research, not evidence that the contribution has already been achieved.

A.7 Future work: a combined Modern AutoCrit and TriAlign tool

Future work item: "somehow integrating modern autocrit and trialign together into one tool." The goal is one application that lets the user choose either component independently or run them together. This is a future integration goal, separate from the current file-based handoff.

Mode	User supplies	Intended result
Modern AutoCrit only	Trial identifiers, protocol text or PDFs	Extract, structure and review criteria; save/export the reviewed records without running TriAlign.
TriAlign only	Existing reviewed criterion files or saved criterion sets	Compare individual criteria or select two trials for alignment and source-linked relationship explanations, without repeating extraction.
Both together	Trial A and Trial B source inputs	Extract both trials, pause for criterion review, then pass the approved records directly to TriAlign and display the comparison.

Keep the extraction and comparison components independently callable, with a shared, versioned handoff schema for trial/criterion IDs, source versions, parent statements, Boolean context, review state and evidence. The combined mode should reuse saved reviewed records and allow users to resume at a chosen stage. The review checkpoint remains visible before comparison; combining the tools must not silently turn unreviewed extractions into validated inputs. A local application can orchestrate the components and preserve their separate exports and traces without requiring website deployment. Interface design, shared job management and end-to-end validation remain to be implemented.

A.8 Risks, mitigations, and definition of done

Risk	Mitigation and reporting requirement
Insufficient pair labels	Prefer a smaller well-supported benchmark, preserve uncertainty, seek independent annotation of a subset, and report learning curves.
Extraction/normalization errors	Preserve raw evidence and correction lineage; version changes; escalate unsupported reference/context assumptions.
Leakage	Map families before mining; group amendments/templates/variants; restrict fitting, prompt development and labeled retrieval memory to approved partitions.
No fine-tuning or routing gain	Report negative findings and ablations; retain the strongest justified frozen configuration.
Unfair external comparisons	Require a matched output contract or identify the work as an adaptation/architectural comparison; do not combine incomparable native metrics.
Uncalibrated confidence	Label scores accurately; select thresholds on development data; report coverage and error together.
Live/API failure	Start locally with predownloaded models and indexes; use bounded retries for optional online calls, explicit error states, cached replay, and a rehearsed recording fallback.
Deployment cost	Run the demonstration on the presenter's machine and distribute versioned model artifacts through the Hub. Track any training or optional LLM API costs separately from website hosting.
Clinical overinterpretation	Keep criterion-level inference, logical witnesses, and protocol precedent distinct from patient screening, population prevalence and causal conclusions.

Done means another person can reconstruct the approved data release, run the documented workflow, reproduce principal results within stated tolerances, inspect every evaluation case and source, operate the demonstration, and distinguish source evidence, model proposals, human adjudication, and unresolved outcomes. Completing the interface or producing a high aggregate score alone does not satisfy that definition.

B Pair labeling contract 0.2

Guide ID: `pair-guide-0.2`. Revised November 7, 2026 from pilot error analysis. Status: operational development guide used by the frozen review in Section C; independent clinical validation remains pending. Corpus: frozen `trial100_v1`. Reviewer identity and evidence determine provenance; a saved Excel status alone does not establish human validation. Sampling rules are not label rules. The original 0.1 guide is retained at the historical guide appendix in this document.

B.1 Unit and identity

An example is an ordered pair A/B of reviewed criteria from different trials. Its ordered ID hashes A then B; its group ID hashes sorted criterion IDs. Reversing a pair changes the ordered ID and swaps only the two restrictive labels. Keep both directions, duplicates, variants and related trial families in the same partition. One unordered pair occurs once in this pilot. A and B presentation direction is seeded, not inferred from presumed restrictiveness.

B.2 Review procedure

1. Read each source, excerpt, parent statement, context and eligibility role. Retain linked Boolean conditions. Structured fields assist inspection; they do not override source evidence. Record a source/structure conflict.
2. Establish a common clinical frame: entity, measurement, population, temporal anchor/window, reference basis, units and applicable exceptions. Do not invent conversions or assume missing context.
3. Interpret the allowed population under each rule, applying internal negation and inclusion/exclusion role once. A matching threshold alone does not establish equivalence. An atom within a compound exclusion may require its siblings before the allowed set can be interpreted.
4. Assign one label below, supported by evidence and rationale. Use ambiguous if unresolved meaning prevents a defensible relationship. Record internal inconsistency as a reason for escalation rather than exploiting empty-set containment to manufacture a restrictive label.
5. Check the reversed pair mentally. Record assumptions, uncertainty and any unresolved question. Leave the final label blank until a reviewer acts.

Label	Decision
`equivalent`	Same allowed set under the supported common frame, including scope and exceptions.
`a_more_restrictive`	A's allowed set is a proper subset of B's.
`b_more_restrictive`	B's allowed set is a proper subset of A's.
`partial_overlap`	Nonempty intersection; neither contains the other.
`contradictory_or_disjoint`	Empty intersection in the shared frame; this is not a whole-trial exclusion claim.
`related_not_comparable`	Clinical relationship is understood but a supported common representation/mapping is unavailable.
`unrelated`	No meaningful direct clinical comparison for this task.
`ambiguous`	Missing, conflicting or uninterpretable evidence prevents a defensible decision.

Examples: inclusion age >=21 is more restrictive than inclusion age >=18. Inclusion age >=18 and exclusion age >=18 are not equivalent. Inclusion age 18–65 and 50–80 partially overlap when all endpoints are inclusive. Different renal measurements may be related but not comparable without a supported mapping. Never infer a negative label from a random sampling designation.

B.3 Annotation unit and inherited context

Record the comparison frame before choosing a label. Name the selected concept, applicable population, quantity/reference basis, temporal anchor and relevant exceptions. Distinguish an atom from its parent or full trial.

- **Independent inclusion conjuncts:** an `AND` parent requires every child. Compare a selected platelet atom without adding unrelated hemoglobin or neutrophil requirements. Preserve qualifiers that actually govern platelets, such as sampling windows or transfusion restrictions.
- **Inclusion alternatives:** one branch of an `OR` need not hold for every eligible participant. Retain the alternative group or explicitly establish a branch-specific frame. Do not make every alternative mandatory.
- **Exclusion alternatives:** when any listed event independently excludes, compare a selected event using its absence as the allowed predicate. Keep governing lookbacks, severity gates and investigator qualifications.
- **Joint exclusions:** exclusion of `AANDB` does not independently exclude A or B. Preserve the conjunction before complementing the condition.
- **Conditional exceptions:** a child such as "for Gilbert's, maximum 2.5 ULN" is not a universal bilirubin rule. Identify its applicability and ordinary branch. Do not silently replace the child with its parent, or narrow both records to a subgroup merely to obtain a label.
- **Permissions:** "patients taking clopidogrel are eligible" permits an exposure; it does not require every eligible participant to take it.

If a child cannot be interpreted independently, record its governing group and evidence. If the stored evidence does not establish the intended unit, use `ambiguous` with reason `scope_unresolved`. A later group-level comparison is a separately identified example, not a silent rewrite of the pair.

B.4 Context rules exposed by the pilot

Issue	Required treatment
Same number, different anchors	Keep screening, consent, randomization, registration and treatment start distinct. Align historical windows only with supporting evidence.
Unstated measurement time	Do not invent a current measurement or equate it with a specified dated sample. When comparing abstract measurement predicates, explicitly limit the frame and do not claim unobserved temporal equivalence.
ULN versus absolute concentration	Require the applicable reference value and assay context for conversion. Never substitute a typical ULN.
Equivalent unit spelling	Supported exact dimensional conversion is allowed. State it and retain strict/inclusive endpoints. Count denominators come from source text.
Unspecified exception limit	Do not interpret "unless due to Gilbert's" or leukemic involvement as unlimited acceptance at every value. Missing replacement policy can make the pair ambiguous.
Consent capability versus completion	Ability, willingness, literacy, signing, approval and signing time are separate predicates. A common entity name does not establish equivalence.
Open-ended illness list	Establish whether listed examples independently exclude and which governing qualifiers apply. Do not assume an exhaustive set.
Bare disease/drug fragment	"Leptomeningeal disease" or "Chemotherapy" may be a heading or assessment-list item. An inclusion flag alone does not establish a required positive predicate.
Malformed source or source/structure conflict	Preserve the source, record the problem and escalate. A normalized comparator cannot silently repair source text.

Issue	Required treatment
Ontology/generalization claim	Omission of "hypertrophic" alone does not prove a broader cardiomyopathy category. Require a supported definition/mapping.

A conditional numerical result belongs in the rationale, for example "B would be stricter on the same sample." It becomes the pair label only when the common-frame condition is supported. Assumptions must be explicit, evidence-supported and accepted by the human adjudicator. They cannot invent missing facts to force an answer. Cite external references and their role; general clinical knowledge does not establish an undocumented protocol policy.

B.5 Choosing and supporting a label

Use `ambiguous` when missing/conflicting **meaning or scope of an input** prevents a defensible choice: malformed text, uncertain polarity, missing heading, undefined exception or unresolved child/parent scope.

Use `related_not_comparable` when requirements are understood but a supported **common operational mapping** is unavailable: absolute creatinine versus a ULN multiple without a reference, or known but unaligned historical windows. Do not use either disposition merely because a case is difficult. Explain what prevents comparison and what evidence would resolve it.

Keep the label separate from workflow status. A human may adjudicate `ambiguous` as the correct final disposition of an incomplete source. A confident proposed `equivalent` can still need adjudication.

For equality/containment, record complete supported predicates. For proper containment, identify a case admitted only by the broader rule. For partial overlap, identify a shared case and one admitted only by each side, retaining parent qualifiers. If witnesses require unknown facts, the relationship is not established. Opposite negation flags alone do not prove disjointness.

B.6 Pilot construction

The original pilot builder, `scripts/build_pair_pilot.py`, read the frozen Parquet and manifest, verified the corpus hash, checked reviewed statuses and unique IDs, and refused to overwrite an existing output directory. That builder is absent from the current checkout; restore a versioned generator before constructing a new pilot. The original JSONL and workbook remain the review artifacts.

The seed is 20261104. Twenty of the 100 trials form a provisional development pool. No final train/validation/test allocation is claimed. Family/duplicate audits were pending at pilot construction; the initial evidence audit is now documented in the trial-family audit appendix in this document, with proposed grouping and explicit unresolved-parentage exclusions. Reserve all twenty trials and related families from future held-out testing. Pilot exposure is development use even if a case is not ultimately labeled. The reservation list lives in the pilot manifest and must be consumed when constructing the eventual family split.

The pilot contains 75 natural cross-trial pairs:

- 25 literal entity matches with differences in reviewed structured fields;
- 25 further literal entity matches;
- 15 source-token lexical neighbors within a matching domain;
- 10 random controls.

Entity matching normalizes case and whitespace, not medical synonyms. Lexical candidates use token Jaccard >=0.25 and rank by that value within their sampling bucket. The original heuristic retains ASCII comparison operators and numbers, but not Unicode inequality symbols as standalone tokens. This sampling limitation does not authorize dropping symbols during labeling. Structured differences are inspection hints, not proof of a relationship. Maximum reuse is three pairs per criterion and four per unordered trial pair. All qualifying sampling memberships are preserved, alongside the method that filled the quota; these are eligibility memberships, not independent sampling probabilities. This enriched convenience sample cannot estimate corpus-wide label

prevalence or support held-out performance claims. No synthetic edits, embedding neighbors, trained predictions or clinical labels are generated.

B.7 Review and versioning

Review the pilot labeling workbook using this document's clinical definitions. Sampling hints and lexical scores are omitted from that packet to reduce anchoring. The machine-readable `pairs.jsonl` retains them, full A/B evidence and originally blank review objects. Save reviewer identity, common frame, cited evidence, rationale, assumptions, ambiguity reason, uncertainty and status (`unreviewed`, `needs_adjudication`, or `adjudicated`). Record ordinal uncertainty as low/medium/high; it is a reviewer report, not a calibrated model probability. An adjudicated record requires a final label and reviewer identity. If there are two reviewers, retain independent decisions and disagreements separately before adjudication rather than overwriting them.

The November 7 AI-assisted workbook pass is a proposed annotation round, not independent human review or gold truth. Preserve that round in `annotations_ai_draft_2026-11-07.jsonl`. In the workbook, the owner can accept, correct or reject its labels, replace the reviewer identity with the actual human reviewer and record the adjudicator/date. Changing status to `adjudicated` without reviewing its evidence does not complete validation. Do not overwrite the AI round to manufacture reviewer agreement.

Before freezing a validated annotation release, require:

- Ordered/group IDs and corpus hashes reconcile to the immutable packet.
- Every final label is in the eight-class vocabulary and has its own frame, evidence, rationale, human reviewer, adjudication date and uncertainty.
- Remaining uncertainty is explicit; source problems have reason codes such as `source_conflict`, `scope_unresolved`, `time_unaligned`, `reference_missing`, `exception_undefined` or `ontology_mapping_missing`.
- Directional cases have a valid witness and pass the A/B swap check.
- Family/split references reconcile to the current audit and reservation map.
- Only human-adjudicated records enter supervised training or clinical evaluation; unresolved review items remain in the review pool.

Existing pilot IDs retain construction guide 0.1. Record guide 0.2 separately on decisions reviewed under this revision; do not relabel the construction packet or silently migrate earlier decisions. A validated release must name the annotation guide actually used and record changes from the AI proposals.

Treat the generated files as an immutable sampling release. Save decisions as a separate versioned annotation file keyed by ordered pair ID, using the review object as a template, so manifest checksums continue to identify the original packet. Do not modify source criteria through a pair decision. Source corrections require a new corpus release and explicit pair migration.

After reviewing the pilot, revise confusing definitions and log contract changes. Audit family/duplicate mappings, reserve development exposures, and freeze the remaining family splits before full mining. Add the teammate's embedding candidates only after input/output and revision checks pass. Keep natural, constructed, development and held-out results distinguishable.

B.8 Scale-up gate and family handling

Treat the family audit as an evidence-backed mapping, not a similarity score. Protocol versions and duplicate registrations belong together. A documented parent/cohort, continued-access, rollover or participant follow-up link can justify a shared split group. Shared sponsor, disease, drug or funding alone does not establish a family. Distinguish clinical families from copied-template leakage groups; either may constrain a split, for different recorded reasons.

Before scaling: validate the pilot, settle material guide changes, version the family/duplicate mapping, reserve pilot-exposed trials and linked families, and resolve or quarantine uncertain parentage. Freeze family-level splits before mining. Missing assignments and cross-split pairs must be rejected. Do not use the current generic CLI defaults as a substitute for this gate.

When adding future trials, rerun the family audit against the union of old and new trials before assigning splits. External parent IDs recorded today must propagate development exclusions if those parents enter the corpus later. For repeated criteria across otherwise distinct families, report overlap and apply an explicit deduplication/purging policy. Common age or consent phrases must not collapse the entire corpus into a single clinical family.

C Pair pilot freeze v1

C.1 Release boundary and review outcome

Release `trial100_v1_pair_review_v1` freezes the November 7, 2026 development review. The reviewer and adjudicator are recorded as Codex. The original sampling packet, construction guide 0.1 identifiers, criterion source text and historical annotation round are unchanged. Current decisions name annotation guide 0.2 separately. This freeze makes a reproducible review snapshot available to collaborators; it does not establish independent human clinical validation, approve supervised training targets, or create a held-out clinical benchmark.

All 75 workbook rows reconcile to the decision ledger. The final pass rechecked the ten changed labels, the seven unresolved relations and the additional source-role problem in pair 10. Pair 18 is both a changed label and an unresolved relation, so these sets must not be added as disjoint counts. Sixty-eight relationship decisions are resolved. Pair 10 retains its `unrelated` relation, but its stored inclusion role conflicts with the source exclusion heading. Excluding that record leaves 67 clean development comparisons. The review queue contains seven unresolved relations plus pair 10: eight records in total.

The primary registry eligibility snapshots cover all twenty pilot trials. Their full eligibility texts match the frozen protocol hashes. The final pass uses those saved primary sources and the cited publications, terminology reference and consent guidance; it does not silently substitute newer source criteria. Original corpus review authority remains the owner's manual validation statement, separate from pair-review authorship.

Pair(s)	Reviewed result	Evidence and scope retained
3	Equivalent	Inclusive adult-age thresholds; consent timing is retained, not equated with an unstated date.
4	Partial overlap	Completed consent and explicit literacy/capability have shared and one-sided witnesses.
8	Equivalent	Registry minimum age and the trial publication corroborate the source typo's inclusive eighteen-year meaning.
15, 61	A more restrictive	Preserve B's governing uncontrolled-illness qualifier for unstable angina and symptomatic CHF.
18	Ambiguous	The leukemic-involvement exception has no stated replacement ceiling; measurement timing is also unaligned.
34, 53	B more restrictive	Hemoglobin and LVEF value thresholds only; no collection-window containment claim.
65	A more restrictive	Hemoglobin value thresholds after exact g/L to g/dL conversion; screening timing remains separate.
70	Equivalent	The cited MedGen/SNOMED concept mapping supports the two obstructive-cardiomyopathy names.

C.2 Scope and outstanding source work

Each frozen decision has a `comparison_scope`. In particular, pairs 3, 34, 53 and 65 compare abstract measurement predicates. Their relationship labels do not imply that complete dated eligibility workflows are nested. A downstream serializer, model or analysis must retain the common frame and governing qualifiers; stripping that frame changes the annotation task.

Pair	Action before promotion or migration
10	Correct the stored inclusion/exclusion role in a new corpus release and migrate the pair; the direct unrelatedness judgment is unchanged.
16	Clarify how the complete open-ended illness lists and their examples inherit scope.
18	Obtain the leukemic-involvement exception policy and a supported measurement-time mapping.
24	Replace the chemotherapy section heading with a valid criterion under a new corpus/pair version.
47	Clarify the exception ceiling and choose an independently interpretable annotation unit.
48	Obtain a controlling source for B's strict versus inclusive age-eighteen boundary; the associated publication did not resolve it.
50	Construct a separately identified whole-rule comparison instead of silently replacing the conditional child.
52	Replace the lesion-measurability fragment with a valid criterion under a new corpus/pair version.

The original source corpus is not edited to conceal these issues. A source correction requires a new corpus version and explicit old-to-new criterion and pair migration. Queued records remain available for review, with their current labels and exact reasons preserved.

C.3 Files, integrity checks and collaboration

The selected corpus snapshot, original pilot files, workbook, decision history, primary-source snapshots and family-audit artifacts are explicitly tracked in Git. Other generated data and Excel lock files remain ignored. The new release lives under `data/processed/pair_pilot_v1/release_v1/`:

- `annotations.jsonl`: all 75 decisions, source links, scope, family/split-group references and review dispositions.
- `resolved_pairs.jsonl`: the 67 clean development comparisons.
- `review_queue.jsonl`: the eight excluded cases and required actions.
- `manifest.json`: input/output SHA-256 hashes, sizes, counts, guide fingerprint, review scope, usage gates and family reservations.

The original sampling manifest remains unchanged, including its historical checksum for the retired Markdown review packet now archived in this TeX source. The freeze command verifies the actual retained sampling JSONL; it does not pretend that the absent original sampler can be regenerated.

From a checkout with project dependencies installed:

```
python scripts/freeze_pair_pilot.py --verify
```

The verifier reads Excel without modifying it, reconciles every review cell and identifier to the ledger, checks source/corpus hashes and primary eligibility text, validates label vocabulary, history, pair uniqueness and reuse caps, and checks family coverage and development exclusions. It then reconstructs the three exports in memory and compares their bytes with the frozen release. Missing files, stale evidence, edited workbook decisions or changed dispositions fail verification. The original freeze command without `--verify` refuses to overwrite an existing release; revisions require a new `--output` directory and a documented version.

All twenty exposed pilot trials and the audit's additional quarantined trial remain excluded from held-out testing: 21 current trials and 841 criteria. Family grouping is still proposed, and no final train/validation/test assignment is claimed. The generic split CLI does not enforce this map. Before scaling into a training or evaluation release, validate the pilot's clinical decisions, settle the guide and source migrations, and freeze an explicit family-aware partition policy. A Git push publishes this development review snapshot and its evidence; it does not change those scientific gates.

D Trial-family audit v1

Prepared November 7, 2026. Evidence audit by Codex; grouping is proposed, not a human-approved split release. Your ongoing workbook validation is untouched.

D.1 Outcome

All 100 frozen trials were reconciled to the upstream cohort and archived eligibility sources. The cohort manifest confirms 50 oncology and 50 cardiovascular trials. Public registry metadata was retrieved for all 100, plus three external records needed to verify family/overlap links.

- 4,950 distinct in-cohort trial pairs screened.
- No shared frozen eligibility-text hash, identical normalized eligibility text, matching organization/protocol identity or explicit in-cohort NCT lineage link found by these checks.
- The union of the top 20 whole-text similarities and top five rare-phrase overlaps yielded 22 unique pairs for evidence review. No reviewed pair supplied sufficient evidence for a family merge.
- Three source-supported links to external trials were recorded, including one participant-follow-up link that constrains splits without establishing identity of the underlying clinical protocols.
- Two trials have unnamed linked/parent protocols and remain quarantined from final held-out testing until resolved or conservatively excluded.
- Current map: 100 provisional clinical families and 100 split groups among the current 100 trials. These are not 100 proven independent families.
- Preserve the 20 pilot-development trial exclusions. Adding the unresolved rollover study excludes 21 current trials / 841 criteria from final held-out testing. Exclusions propagate through known family/participant links.

The current corpus has one observed registration per proposed family. The external anchors matter when related records enter the corpus later.

D.2 Source-supported external links

Current trial	Linked record	Evidence and treatment
NCT01760291, CAP2	NCT01182441, PREVAIL	CAP2 explicitly describes continued access during PREVAIL analysis/review. Share a clinical family and split group. The pilot-development exclusion propagates to PREVAIL.
NCT04880434, ZUMA-2 cohort 3	NCT02601313, ZUMA-2 cohorts 1/2	The description explicitly identifies separate registrations of KTE-C19-102 cohorts. Matching protocol/EudraCT information supports the clinical family and split constraint.
NCT04880434	NCT05041309, Kite long-term follow-up	The description transfers remaining follow-up to this study. Record participant overlap and a shared split group. Do not claim all Kite treatment protocols are one clinical family.

Sources: CAP2 registry record, PREVAIL, ZUMA-2 cohort 3, cohorts 1/2, and long-term follow-up. The downloaded snapshots preserve the evidence used for these decisions.

D.3 Unresolved parentage and conservative exclusions

Trial	Unresolved item	Action
NCT05419375, BX43361	Master screening for linked Roche studies; the screened records do not enumerate the linked trial IDs.	Already pilot-reserved. Retain held-out exclusion and identify linked protocols before adding them.
NCT07172919, protocol 20250089	Rollover from Amgen sotorasib parent studies; parent IDs are not enumerated in the screened records.	Add held-out quarantine. Do not infer named parents merely from the drug name.

Sources: master-screening record and rollover record. These unresolved links do not require delaying development-only labeling; they do prevent using those records as supposedly independent held-out cases.

D.4 Rejected linkage shortcuts

Five shared secondary-identifier groups were found. U10CA031946 is shared by NCT00025207 and NCT01841736, but shared grant/cooperative-group support does not establish one protocol. Four other shared identifiers are year-like entries: 2016, 2018, 2023 and 2025. None is used as a family key.

The strongest whole-eligibility similarity was NIREUS versus HARMONEE: cosine 0.303. Registry evidence describes different sponsors, protocol IDs and stent interventions. The strongest rare-phrase coverage was 13.0% for NCT00025207 versus NCT00028587, which have different NCI identifiers, treatments and disease scopes. Neither similarity alone establishes a family.

All 22 review decisions, with source URLs and rationales, are preserved in `configs/family_audit_decisions_v1.json`. `no_merge_evidence` means no supported merge at this screening stage, not proof that no relationship exists.

D.5 Method and limitations

1. Verify the frozen Parquet hash and reconcile all trial IDs to the upstream cohort. For each trial, load its source job(s) and reconcile source hashes. In this NCT ingestion workflow, the field called `protocol_hash` hashes ingested eligibility text, not an independently audited full protocol PDF.
2. Compare current registry organization/protocol IDs, NCT aliases and explicit NCT references in archived eligibility and registry descriptions. Retain secondary-ID matches as evidence requiring interpretation, not automatic family merges. Public identifiers have different meanings: see the registry definitions.
3. Screen full archived eligibility text with TF-IDF word unigrams/bigrams, rare exact reviewed-parent matches and rare eight-word phrases. Exact parent matching retains written operators. Word-based heuristics retain numbers but do not establish comparator equivalence. Exclude widely repeated phrases from rare matching.
4. Fixed strong-signal thresholds: whole-text cosine >=0.60; at least three shared parent statements of >=12 tokens occurring in 2–5 trials; or >=20 rare eight-word matches covering >=20% of the shorter shingle set. No pair crossed these thresholds. They are declared heuristics, not validated sensitivity guarantees. The ranked review queue supplements them.
5. Review metadata/wording evidence for the 22 prioritized pairs and verify named external records. Separate family identity, participant overlap, template similarity and insufficient evidence.
6. Build clinical-family and split-group components separately. Propagate pilot-development exclusions and unresolved-parentage quarantines across split groups. Do not assign train/validation/test partitions in this audit.

Registry metadata retrieved November 7 may postdate frozen source versions; it is lineage evidence, not a replacement for frozen criterion text. The cohort's complete public protocol PDFs and all unnamed external

parents have not been exhaustively audited. Near-duplicates below the heuristic thresholds can remain. Audit similarities must not be presented as clinical performance metrics or reused as a trained retrieval baseline: this vectorizer serves pre-split grouping review only.

Common age, consent or laboratory clauses do not merge all trials into a family. The eventual evaluation also needs a separate criterion/pair overlap policy: report repeated content and purge cross-partition duplicates according to the intended generalization claim. Trial-family grouping alone cannot remove all criterion-level overlap.

D.6 Artifacts and reproduction

Source code:

- `scripts/audit_trial_families.py`: coverage, provenance and pairwise screen.
- `configs/family_audit_decisions_v1.json`: explicit evidence-review ledger.
- `scripts/finalize_family_audit.py`: complete maps and exclusion propagation.

Generated artifacts live in `data/processed/family_audit_v1/`:

- `registry_snapshot.json` and `external_registry_snapshot.json`: relevant public metadata with retrieval timestamps, omitting contacts.
- `trial_inventory.json`: all 100 trials, frozen text/source hashes and metadata.
- `all_pair_scores.json`: all 4,950 screening pairs.
- `candidate_edges.json`: 22 prioritized pairs.
- `shared_secondary_identifiers.json`: five rejected shortcut groups.
- `screening_manifest.json`: inputs, thresholds, hashes and coverage counts.
- `family_mapping.json`: 100 current trial assignments, external nodes, `family_by_trial`, `split_group_by_trial` and held-out exclusions. It contains no final partition assignments.

Generated data are ignored by Git under the existing project policy. Preserve the snapshot files with the audit release; the source ledger alone cannot reproduce metadata that changes remotely.

Recompute the offline audit from its saved snapshots, from the repository root in the project environment:

```
python scripts/audit_trial_families.py \
  --cohort /path/to/exported/cohort.json \
  --jobs /path/to/exported/source_jobs
python scripts/finalize_family_audit.py
```

Audit regeneration requires explicit `--jobs` and `--cohort` paths to exported files, with no assumed sibling checkout. Normal demo execution uses the bundled inventory and does not run this script. It needs the archived upstream source jobs plus the frozen corpus and registry snapshot. Fresh registry retrieval requires `--fetch-registry` and a new output directory; existing snapshots cannot be overwritten through that option. The finalizer rejects stale evidence hashes and missing/duplicate decisions.

D.7 Gate before scaling

Finish the human pilot validation under guide 0.2 and freeze the decision release. Then construct a versioned split artifact that consumes **both** `split_group_by_trial` and `holdout_excluded_trial_ids`. Resolve the two unnamed parent sets or keep their conservative exclusions. Verify every criterion's trial has an assignment, reject cross-split pairs, and check that no pilot or linked/quarantined trial reaches the final test partition.

The current generic `trialignsplit` CLI does not consume this map or enforce the reservations, and the generic pair generator does not reject every missing assignment. Do not use those defaults for scale-up. Integrating and validating the audited split contract is the next implementation step after the owner's label review; no final splits or new large pair pool were created prematurely.

E Progress presentation and demonstration plan

Prepared November 6 and updated November 7, 2026. The deck separates implemented prototype behavior from training and clinical evaluation. The local runtime is implemented; presentation-machine rehearsal remains required.

E.1 Speaking and timing

Slides	Speaker	Time
1–5: objective, trial workflow, corpus and relationships	Jonathan	4:00
6–7: agent routing, skills, tools and hooks	Jiatong	2:00
8–10: retrieval, related systems and prototype cases	Jiatong	2:30
Live walkthrough outside the slides	Jiatong, with Jonathan explaining one label decision	3:00
11: limitations	Jonathan	1:00
12: next steps	Both	1:30
Application switching and handoffs	Both	1:00
13: questions and feedback	Both	5:00

Total: 20 minutes. The current deck has 13 slides, including the shared trial-comparison diagram and related-system positioning, with no pair-construction, demo-plan or backup slides. The demonstration happens outside the slide deck. Speaker notes, source footers and engineering-check counts were removed from the deck. Speaking allocations and timing remain in this document. The editable shared diagram is `docs/figures/trial_comparison_workflow.tex`; both documents include that file directly.

E.2 Recommended demonstration: a small local TriAlign review UI

Historical design proposal, retained for context: build a narrow workflow around the existing 75-pair pilot. The implemented trial-comparison runtime is described in Section 5.7. A trained model is not required to demonstrate evidence inspection and durable human annotation.

Minimum behavior:

1. Load the frozen pilot JSONL and identify the selected pair by its stable ID.
2. Display A/B source text, eligibility role and negation side by side.
3. Expand parent statements, context, units, temporal qualifiers and provenance.
4. Let a reviewer enter a label, common frame, rationale, evidence, uncertainty and status. Preserve unresolved cases as needs_adjudication.
5. Validate required fields before accepting an adjudicated label.
6. Save to a separate local annotation store. Restart and reload the decision.

Keep source records and the original sampling release immutable. Record guide and corpus versions alongside each decision. Practice in a separate demo annotation store so rehearsal entries never become gold labels accidentally. Do not connect the current exact comparator to final automatic decisions until the documented polarity/context defect is fixed and validated.

Three-minute sequence:

- 0:00–0:30: choose a prepared pilot pair and show its identity.
- 0:30–1:15: inspect source and parent context, explain the common frame.
- 1:15–2:00: record a human decision with evidence, or explain why it remains unresolved.
- 2:00–2:30: save and reload to show persistence.
- 2:30–3:00: open a second context-sensitive case and show escalation.

Select cases after clinical review. Do not preassign a label solely from sampling method or lexical score. If no pilot decision is ready, demonstrate saving an unresolved review rather than inventing adjudication.

The local browser implementation now supports trial selection, criterion imports, evidence inspection, bounded agent execution and durable review history. Use the launch command in Section 5.8. The earlier narrow pilot-UI proposal above is retained as design history; the current runtime compares trials using frozen criterion inputs.

E.3 Alternative demonstration: Modern AutoCrit

Use the existing upstream application to explain how the reviewed corpus was produced. Its documented workflow accepts NCT IDs, pasted text or PDFs, extracts structured criteria, supports correction, and exports reviewed rows. The README documents local access at `http://127.0.0.1:8765`.

Suggested sequence: open a prepared job, inspect one source-backed row, make a clearly identified practice correction, then show reviewed export. Use a disposable/demo job and avoid changing validated corpus records. Live extraction is optional and introduces API/network uncertainty. Rehearse startup, credentials and a failed-call case before deciding to use it. Never display credentials or unrelated local records on the projector.

This demonstrates upstream extraction, not completed TriAlign relationship prediction. Explain that distinction explicitly. It is the lower-build option only if the actual app runs reliably in rehearsal; that has not been verified as part of this deck task.

E.4 Presentation scope

The owner removed the demo-plan slide and all backup slides. Do not add them back without a new request. The existing workbook image assets remain source artifacts, not evidence of a functioning runtime UI. Runtime implementation is complete at prototype scope; rehearsal and clinical acceptance remain pending. Keep these planning notes in this technical document, not in an additional Markdown plan or in the deck.

E.5 Evidence and reporting boundaries

- Dataset count: the frozen manifest records 3,537 criteria, 100 trials, 3,537 corrected rows and zero recorded issues/blockers. Review authority comes from the owner's manual validation statement, not independent annotation.
- Pilot: manifest records 75 unlabeled natural pairs from a reserved pool of 815 criteria across 20 trials. An initial family audit was completed November 7; its proposed groups and unresolved-parentage exclusions are documented in the audit appendix. The inspected labeling workbook had zero entered labels on November 6.
- Baselines: local teammate branch feature/comparisons-agent-ui at d30c71e contains shared retrieval code and dense wrappers. The 200-criterion sample, 20 passing tests and saved-output checks come from her report. Saved dense outputs are absent here, and this deck does not claim full reproduction.
- Failure case: a constructed pair with identical age >=18 intervals and opposite inclusion/exclusion role returned equivalent with confidence 1.0 from the current exact tool. That is a diagnostic failure, not an error rate.
- Runtime UI and bounded LLM judge/critic are implemented. The trained encoder, calibrated uncertainty and clinical evaluation remain pending. No clinical accuracy or latency benchmark is claimed.

Her branch was inspected read-only using git show. The working checkout remained on main, with no edits to her implementation and no branch switch.

E.6 Build

From the repository root:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  -outdir=/private/tmp/trialign-progress-build docs/progress_presentation.tex
```

Run the presentation build command from the repository root. To display presenter notes, change hide notes to show notes in a separate temporary build. Keep the ordinary audience PDF free of notes.

F Historical pair labeling contract 0.1

Historical construction specification. This appendix preserves the original pilot rules; guide 0.2 governs new adjudication. Historical file paths and executable names below describe the original release, not necessarily current available files.

Status: development pilot; revise after human review. Corpus: frozen `trial100_v1`. The pilot has no proposed or gold labels. Sampling rules are not label rules.

F.1 Unit and identity

An example is an ordered pair A/B of reviewed criteria from different trials. Its ordered ID hashes A then B; its group ID hashes sorted criterion IDs. Reversing a pair changes the ordered ID and swaps only the two restrictive labels. Keep both directions, duplicates, variants and related trial families in the same partition. One unordered pair occurs once in this pilot. A and B presentation direction is seeded, not inferred from presumed restrictiveness.

F.2 Review procedure

1. Read each source, excerpt, parent statement, context and eligibility role. Retain linked Boolean conditions. Structured fields assist inspection; they do not override source evidence. Record a source/structure conflict.
2. Establish a common clinical frame: entity, measurement, population, temporal anchor/window, reference basis, units and applicable exceptions. Do not invent conversions or assume missing context.
3. Interpret the allowed population under each rule, applying internal negation and inclusion/exclusion role once. A matching threshold alone does not establish equivalence. An atom within a compound exclusion may require its siblings before the allowed set can be interpreted.
4. Assign one label below, supported by evidence and rationale. Use ambiguous if unresolved meaning prevents a defensible relationship. Record internal inconsistency as a reason for escalation rather than exploiting empty-set containment to manufacture a restrictive label.
5. Check the reversed pair mentally. Record assumptions, uncertainty and any unresolved question. Leave the final label blank until a reviewer acts.

Label	Decision
`equivalent`	Same allowed set under the supported common frame, including scope and exceptions.
`a_more_restrictive`	A's allowed set is a proper subset of B's.
`b_more_restrictive`	B's allowed set is a proper subset of A's.
`partial_overlap`	Nonempty intersection; neither contains the other.
`contradictory_or_disjoint`	Empty intersection in the shared frame; this is not a whole-trial exclusion claim.

Label	Decision
`related_not_comparable`	Clinical relationship is understood but a supported common representation/mapping is unavailable.
`unrelated`	No meaningful direct clinical comparison for this task.
`ambiguous`	Missing, conflicting or uninterpretable evidence prevents a defensible decision.

Examples: inclusion age >=21 is more restrictive than inclusion age >=18. Inclusion age >=18 and exclusion age >=18 are not equivalent. Inclusion age 18–65 and 50–80 partially overlap when all endpoints are inclusive. Different renal measurements may be related but not comparable without a supported mapping. Never infer a negative label from a random sampling designation.

F.3 Pilot construction

Run `pythonscripts/build_pair_pilot.py` from the repository root in the project environment. It reads the frozen Parquet and manifest, verifies the corpus hash, checks reviewed statuses and unique IDs, and refuses to overwrite an existing output directory. Use `--output` for a new explicitly versioned run.

The seed is 20261104. Twenty of the 100 trials form a provisional development pool. No final train/validation/test allocation is claimed: family and duplicate audits are pending. Reserve all twenty trials and any related families from future held-out testing. Pilot exposure is development use even if a case is not ultimately labeled. The reservation list lives in the pilot manifest and must be consumed when constructing the eventual family split.

The pilot contains 75 natural cross-trial pairs:

- 25 literal entity matches with differences in reviewed structured fields;
- 25 further literal entity matches;
- 15 source-token lexical neighbors within a matching domain;
- 10 random controls.

Entity matching normalizes case and whitespace, not medical synonyms. Lexical candidates use token Jaccard >=0.25 and rank by that value within their sampling bucket; comparison operators and numbers are retained as tokens. Structured differences are inspection hints, not proof of a relationship. Maximum reuse is three pairs per criterion and four per unordered trial pair. All qualifying sampling memberships are preserved, alongside the method that filled the quota; these are eligibility memberships, not independent sampling probabilities. This enriched convenience sample cannot estimate corpus-wide label prevalence or support held-out performance claims. No synthetic edits, embedding neighbors, trained predictions or clinical labels are generated.

F.4 Review and versioning

Review the original pilot review packet, now archived within this TeX source using the technical guide's clinical definitions. Sampling hints and lexical scores are omitted from that packet to reduce anchoring. The machine-readable `pairs.jsonl` retains them, full A/B evidence and blank review objects. Save reviewer identity, common frame, cited evidence, rationale, assumptions, ambiguity reason, uncertainty and status (`unreviewed`, `needs_adjudication`, or `adjudicated`). Record ordinal uncertainty as low/medium/high; it is a reviewer report, not a calibrated model probability. An adjudicated record requires a final label and reviewer identity. If there are two reviewers, retain independent decisions and disagreements separately before adjudication rather than overwriting them.

Treat the generated files as an immutable sampling release. Save decisions as a separate versioned annotation file keyed by ordered pair ID, using the review object as a template, so manifest checksums continue to identify the original packet. Do not modify source criteria through a pair decision. Source corrections require a new corpus release and explicit pair migration.

After reviewing the pilot, revise confusing definitions and log contract changes. Audit family/duplicate mappings, reserve development exposures, and freeze the remaining family splits before full mining. Add the teammate's embedding candidates only after input/output and revision checks pass. Keep natural, constructed, development and held-out results distinguishable.

G Documentation policy, migration and change history

This document is the single source for project research notes, labeling contracts, family audits, demonstration planning and substantive change history. Keep the main README and Python/R Git guides as the only standalone Markdown documentation. The README provides setup and pointers; put future methodology, implementation decisions and detailed workflow notes here. Presentation sources remain separate deliverables. JSON, Parquet, workbooks, source code and machine-readable audit ledgers remain data/code artifacts.

The former Markdown contracts, audit, demonstration notes and empty changelog have been consolidated into the appendices above. The original changelog contained only empty Added, Changed and Deleted headings, with no historical entries to migrate. The November 7 consolidation creates this policy and replaces stale references in the README and Git guides.

The original 75-pair Markdown packet duplicated source evidence already retained in the pilot JSONL and workbook. Use those artifacts for review; the complete original packet is preserved, byte for byte, in the embedded source archive below rather than repeated as a second 75-pair printout. The original pilot manifest still records its historical checksum. Its absence as a loose Markdown file is a documentation migration, not a change to the sampling release. Validators reconstructing that historical artifact must resolve it from this archive before checking the original checksum.

Every retired Markdown file is preserved in the TeX comments as a base64 archive with its original path, byte length and SHA-256. This retains the historical wording, including superseded plans, without treating it as the current operating instructions. Recovery and integrity verification use only the Python standard library:

```
import base64, hashlib, json, re
from pathlib import Path
source = Path("docs/trialign_technical.tex").read_text()
pattern = r"(?m)^% ARCHIVE_BEGIN ([^\n]+)\n(.*?)^% ARCHIVE_END$"
for match in re.finditer(pattern, source, re.S):
    info = json.loads(match[1])
    encoded = "".join(line[2:] for line in match[2].splitlines())
    original = base64.b64decode(encoded)
    assert len(original) == info["bytes"]
    assert hashlib.sha256(original).hexdigest() == info["sha256"]
    print(info["path"], info["sha256"])
```

Recovery does not write new Markdown files unless the caller explicitly chooses to materialize an original artifact. The operational content remains readable in this document; the historical packet data also remain in Excel and JSONL. Do not create new Markdown notes or changelogs for future tasks.

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